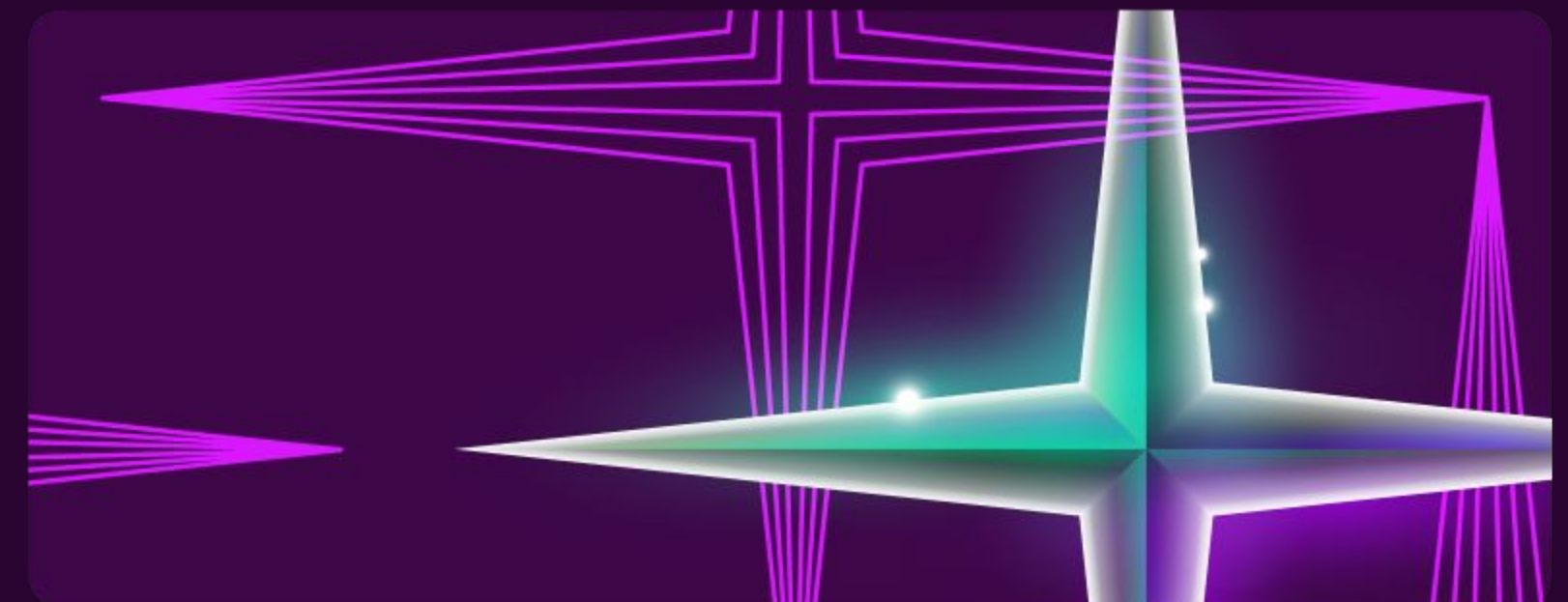
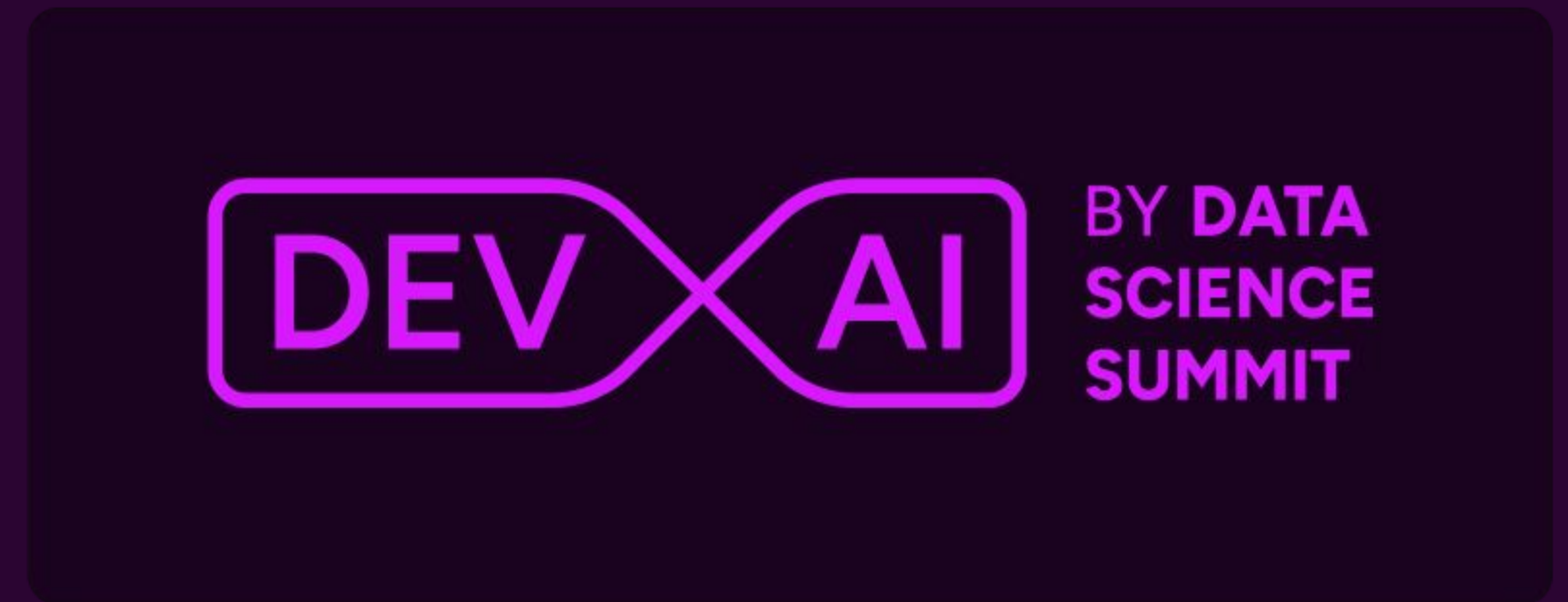


# Unifying Language Models with Knowledge Graphs

A Seshaditya

naavi network



- **Special thanks to**

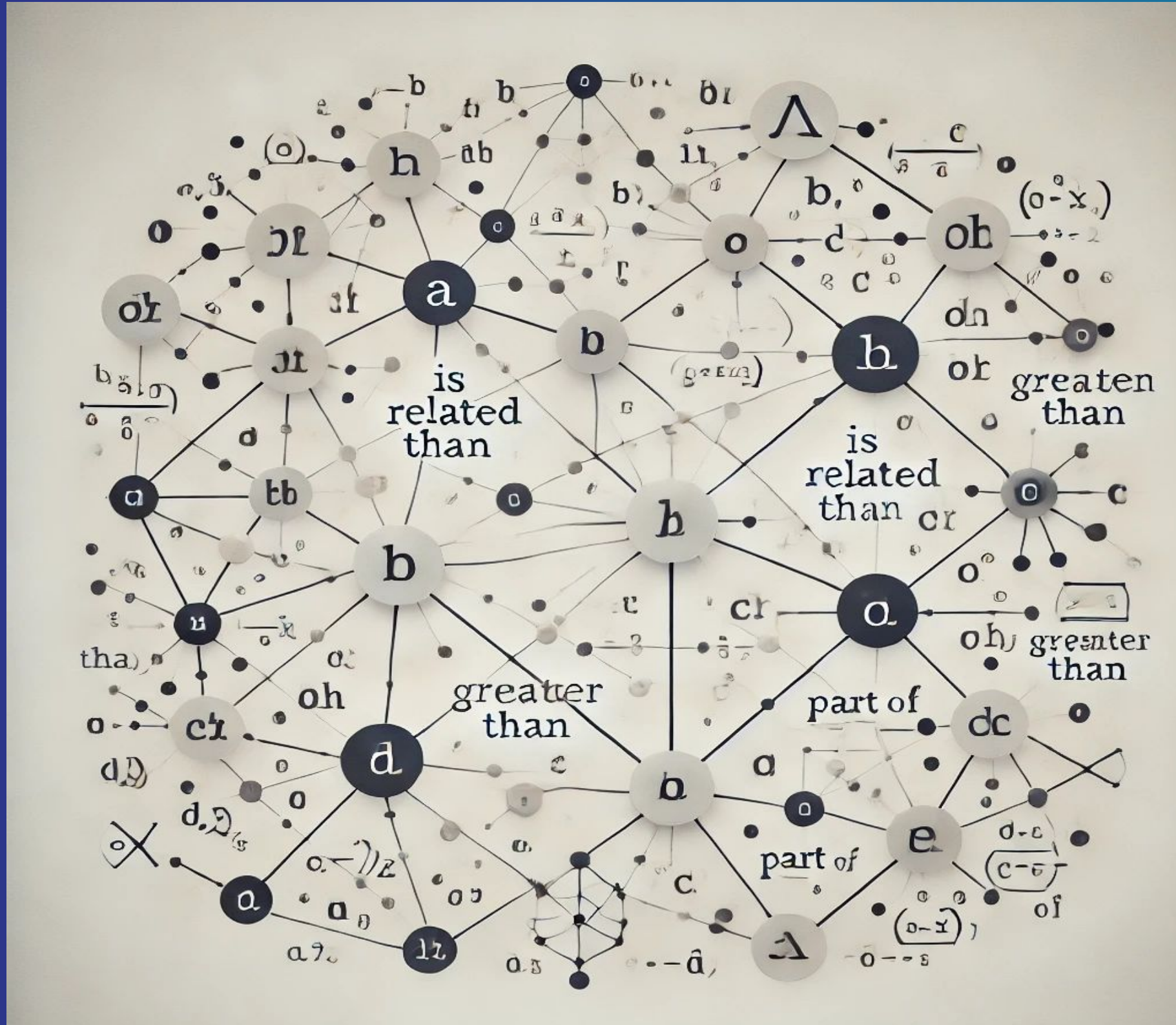
- **Jakob Porschmann (Google Berlin)**
- **Anuradha**
- **Lavanya**
- **Siva Sai**

# Table of Contents

- **Knowledge Graphs**
- **Unifying Language models with Knowledge Graphs (KG)**
  - **KG enhanced language Model generation**
  - **Language Models augmented KG**
  - **Synergising Language Models with KG**
- **Evaluation and Benchmarking**
- **Generating Personalised educational Pathways**



# Knowledge Graphs



# Information as entities and relationship

## Key Component

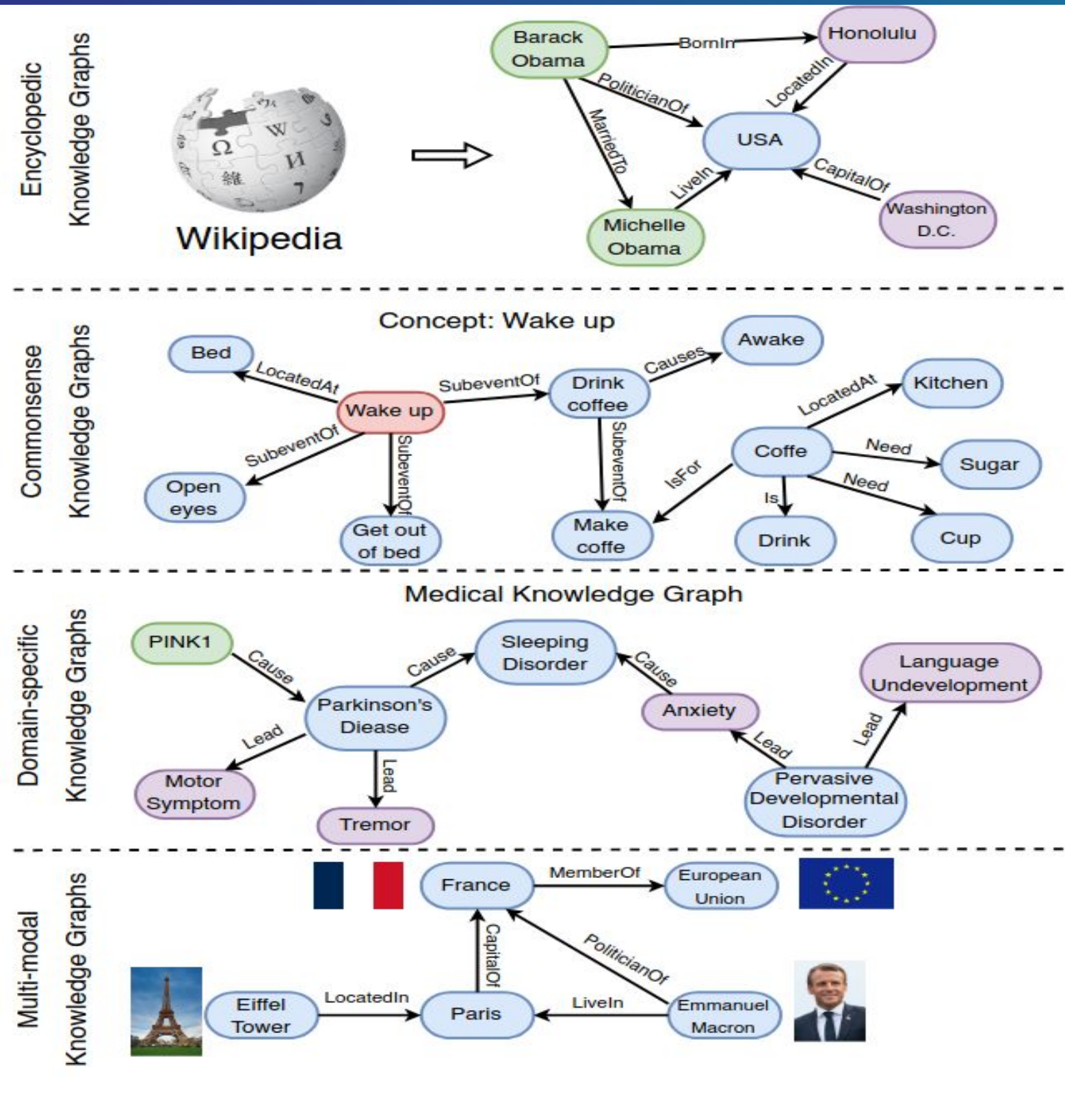
# Nodes

# Edges

## Attributes

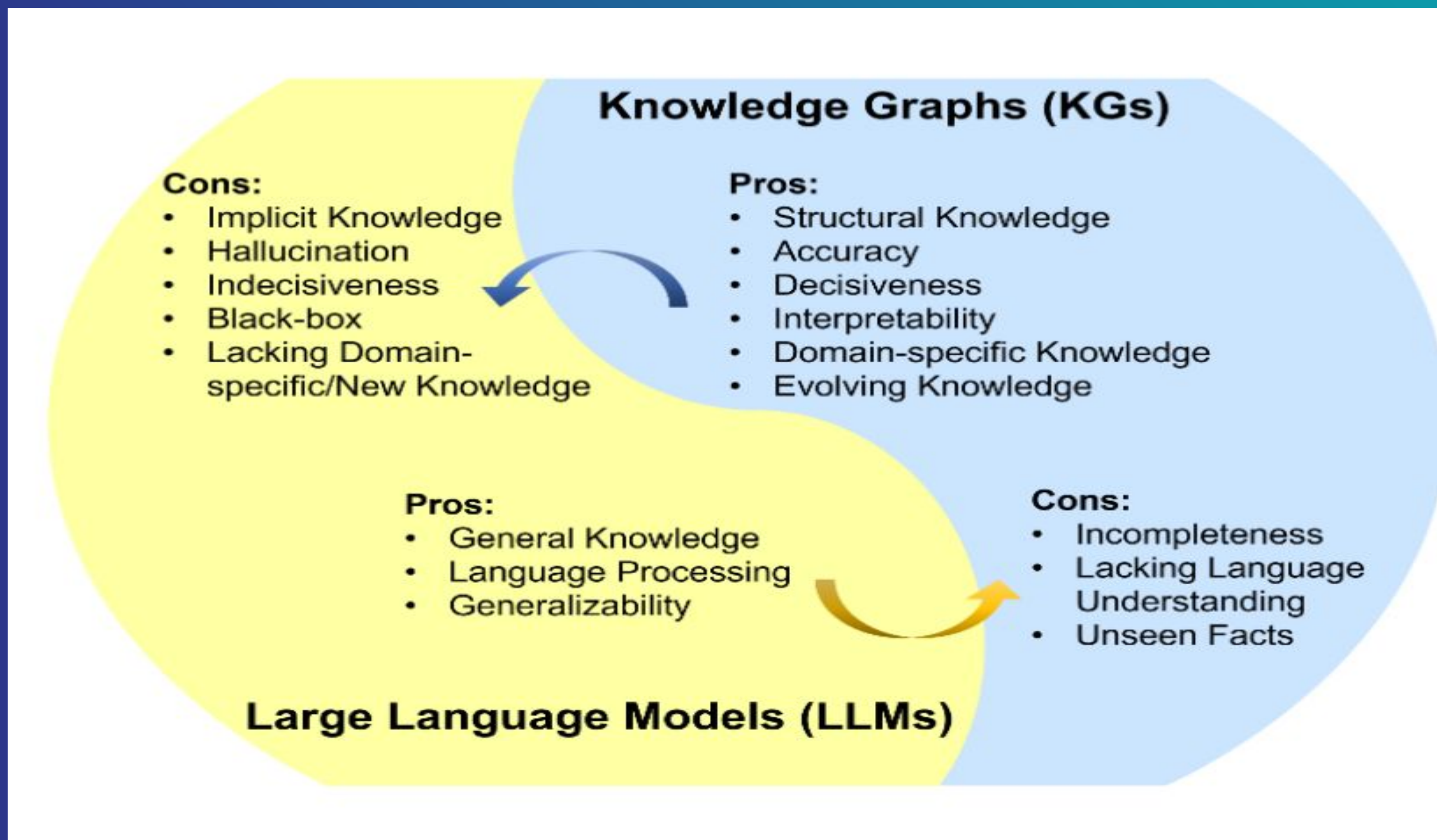


# Different Knowledge Graphs



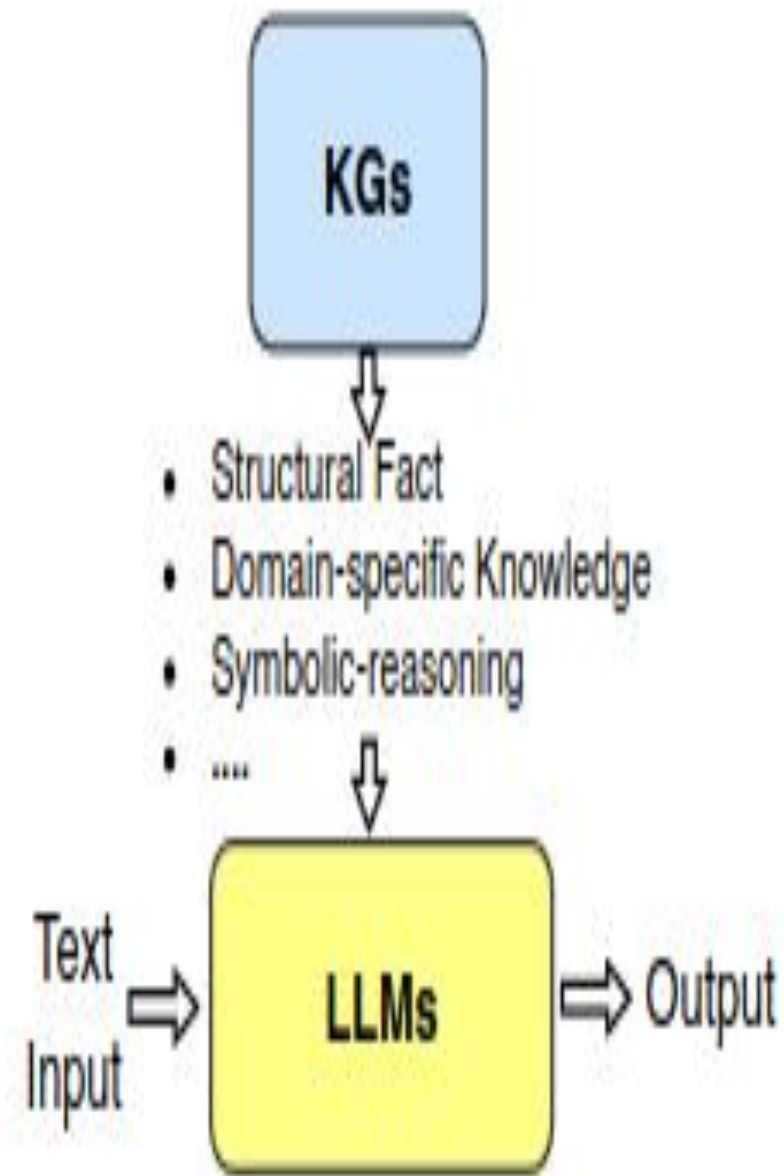
- Encyclopedic Knowledge Graphs
- Commonsense Knowledge Graphs
- Domain-specific Knowledge Graphs
- Multi-modal Knowledge Graphs

# LLMs and Knowledge Graphs

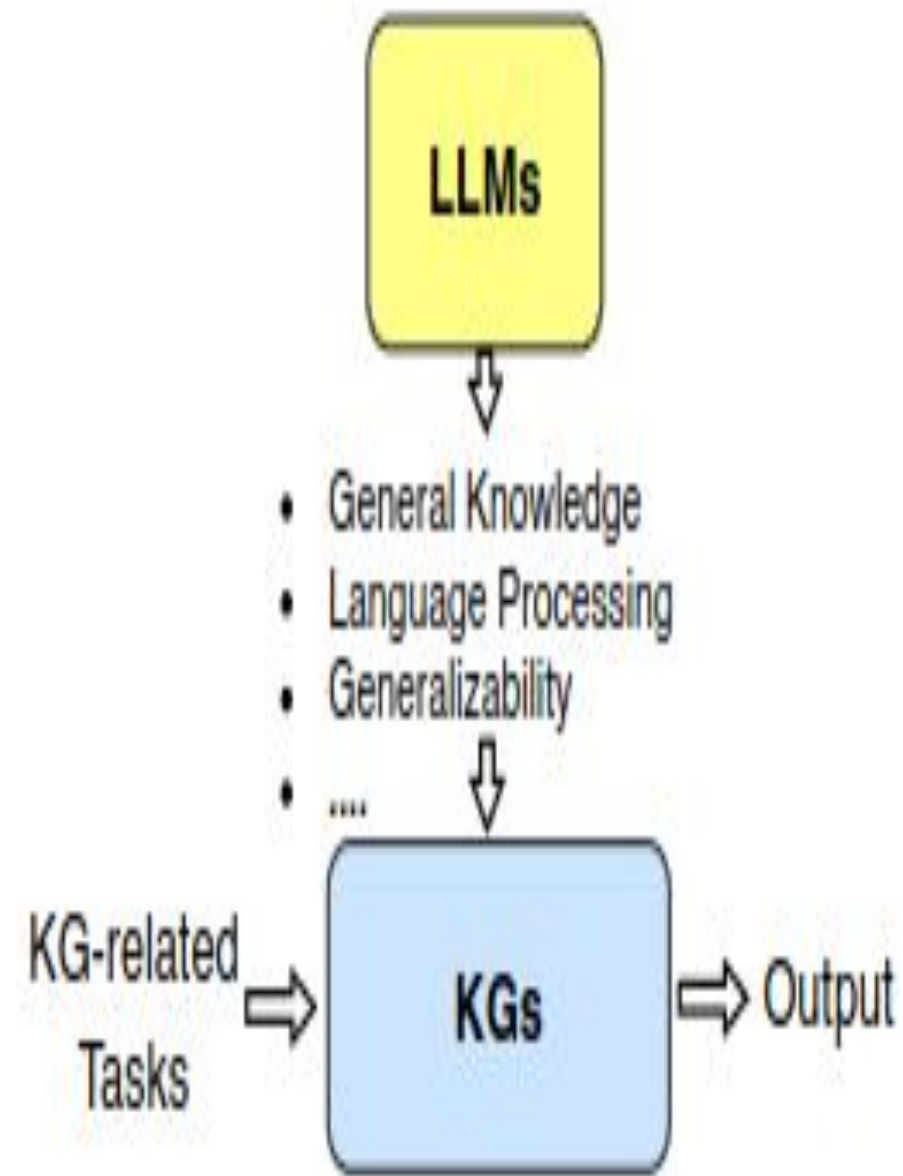




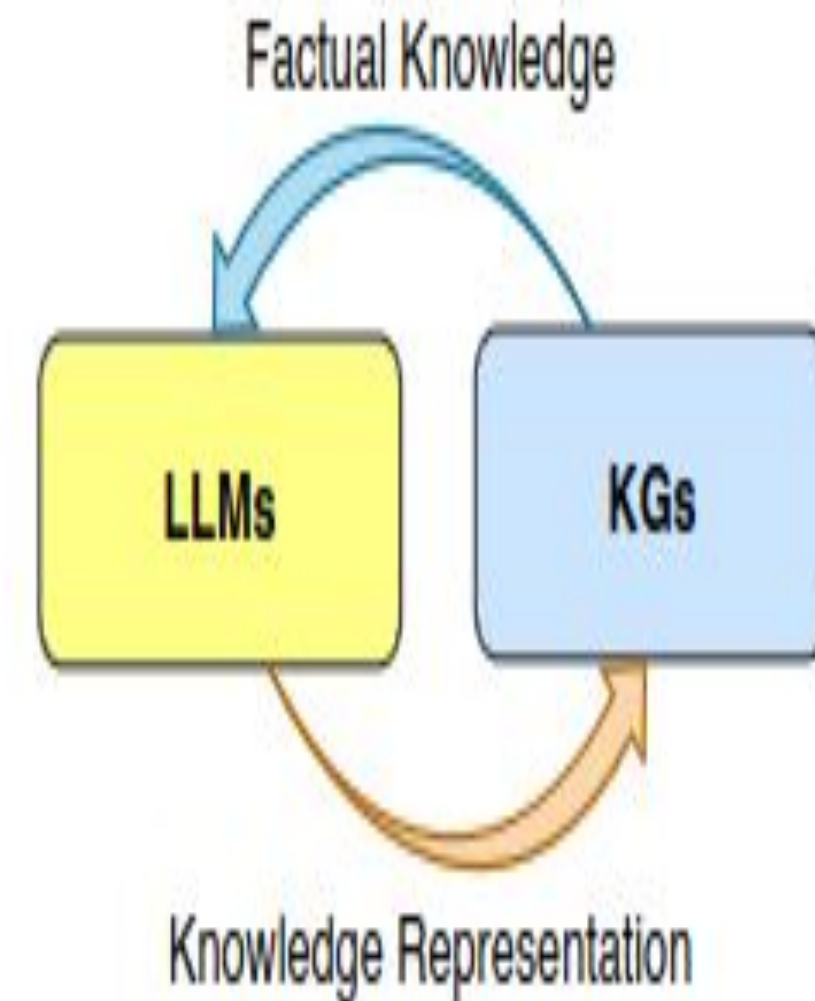
# Different approaches



a. KG-enhanced LLMs

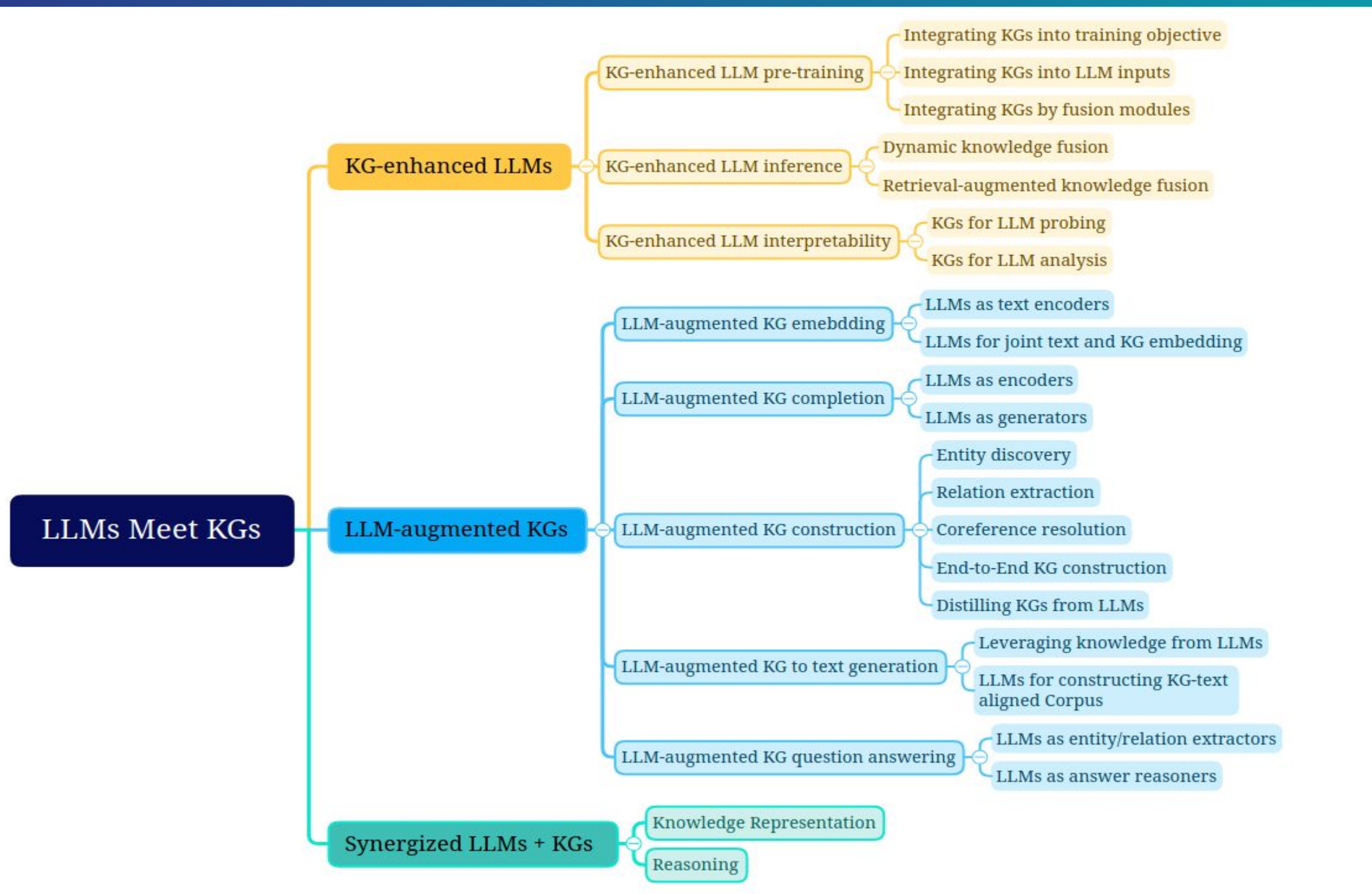


b. LLM-augmented KGs



c. Synergized LLMs + KGs

# Unifying LMs with Knowledge Graphs





# Knowledge Graph enhanced LLM generation

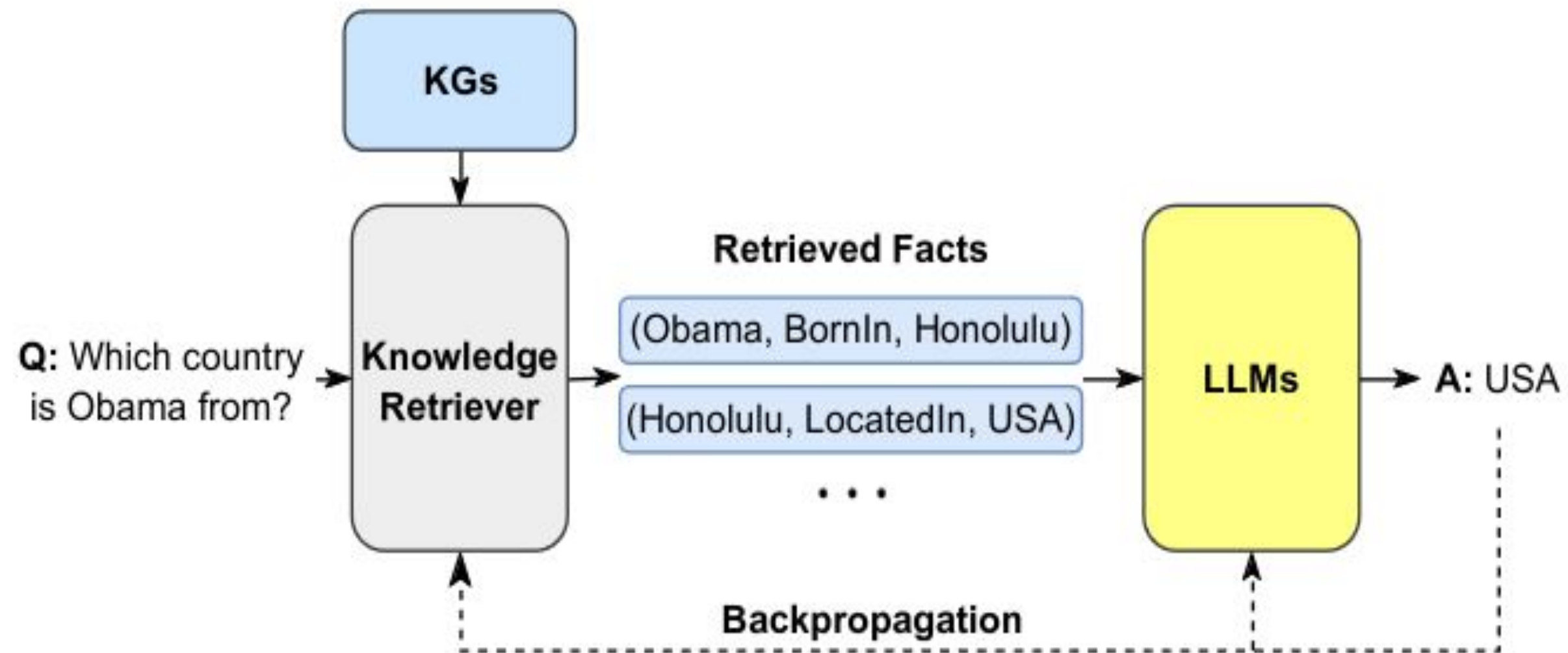
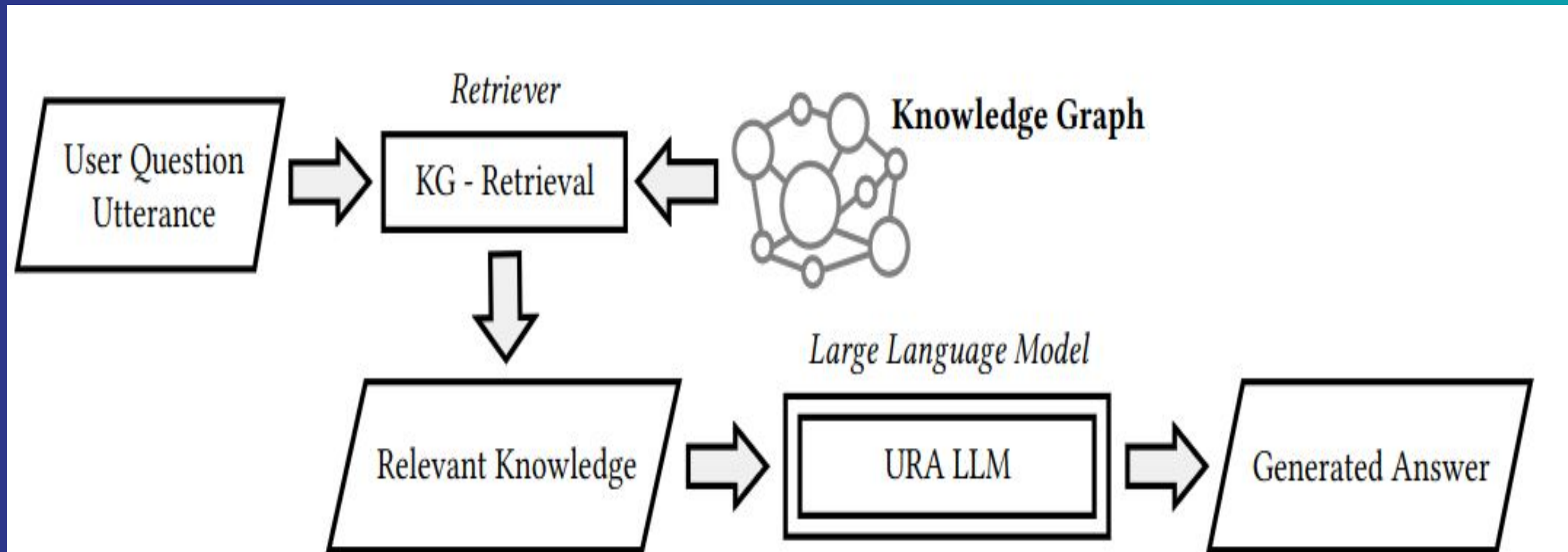


Fig. 11. Retrieving external knowledge to enhance the LLM generation.

# Knowledge Graph enhanced LLM generation



(a) KG-augmented LLMs Approach



# Knowledge Graph enhanced LLM generation

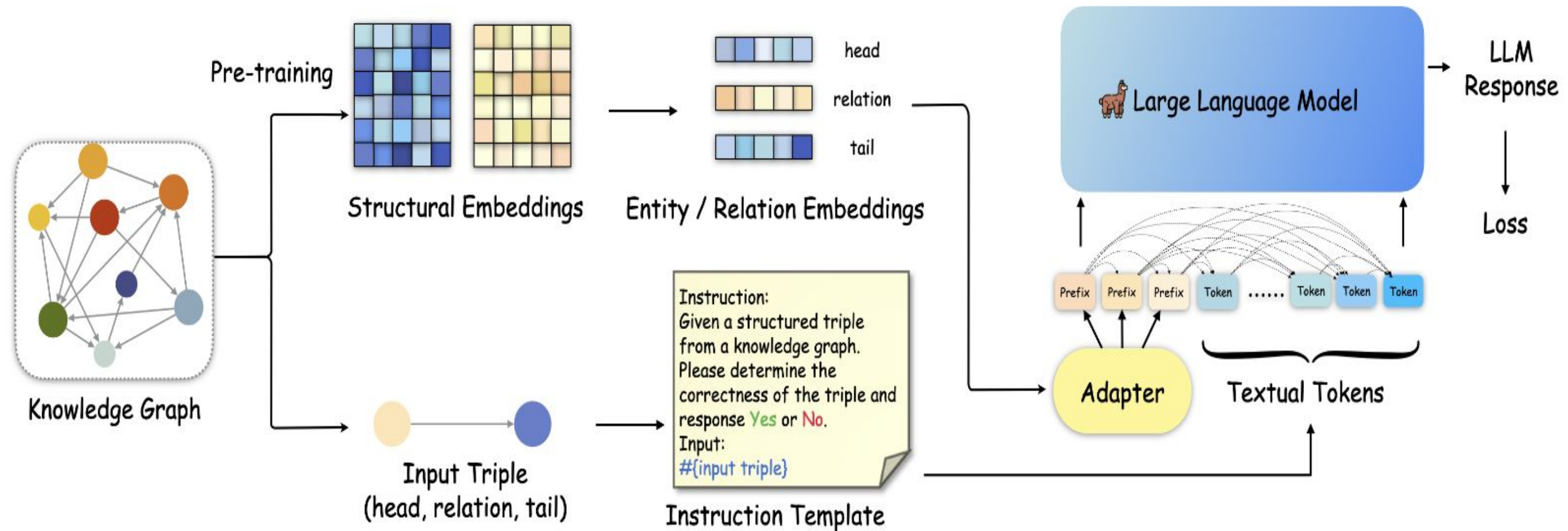
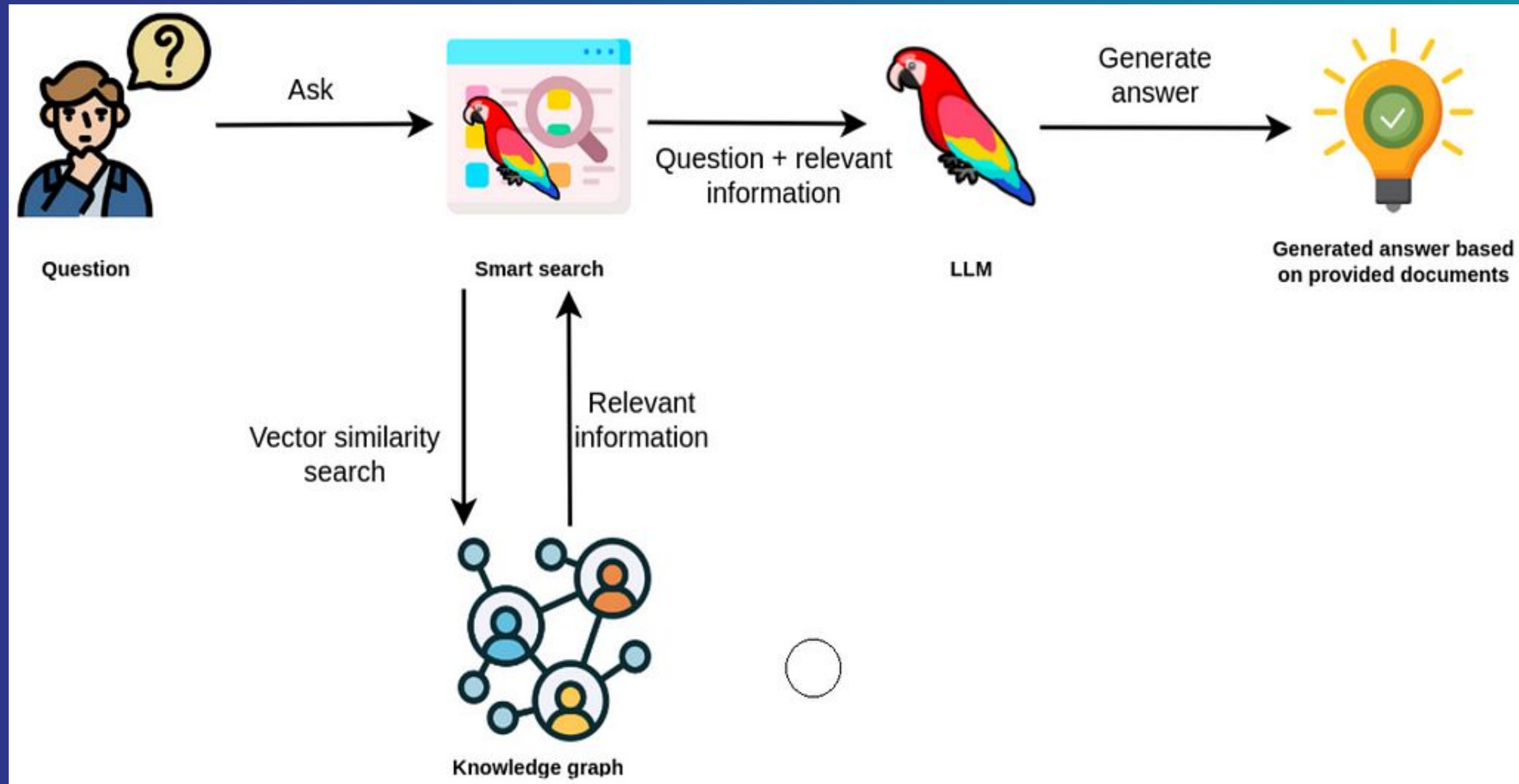


Figure 2: An overview of the knowledge prefix adapter (KoPA) by us. KoPA first pre-trains structural embeddings for the entities and relations in the given KG and then employs instruction tuning to fine-tune the LLM. The structural embeddings of the given input triple will be projected into the textual space of the LLM by the adapter and serve as prefix tokens in the front of the input sequence, which can be "seen" by the following textual tokens due to the unidirectional attention mechanism in the decoder-only LLM.

# Graph RAGs

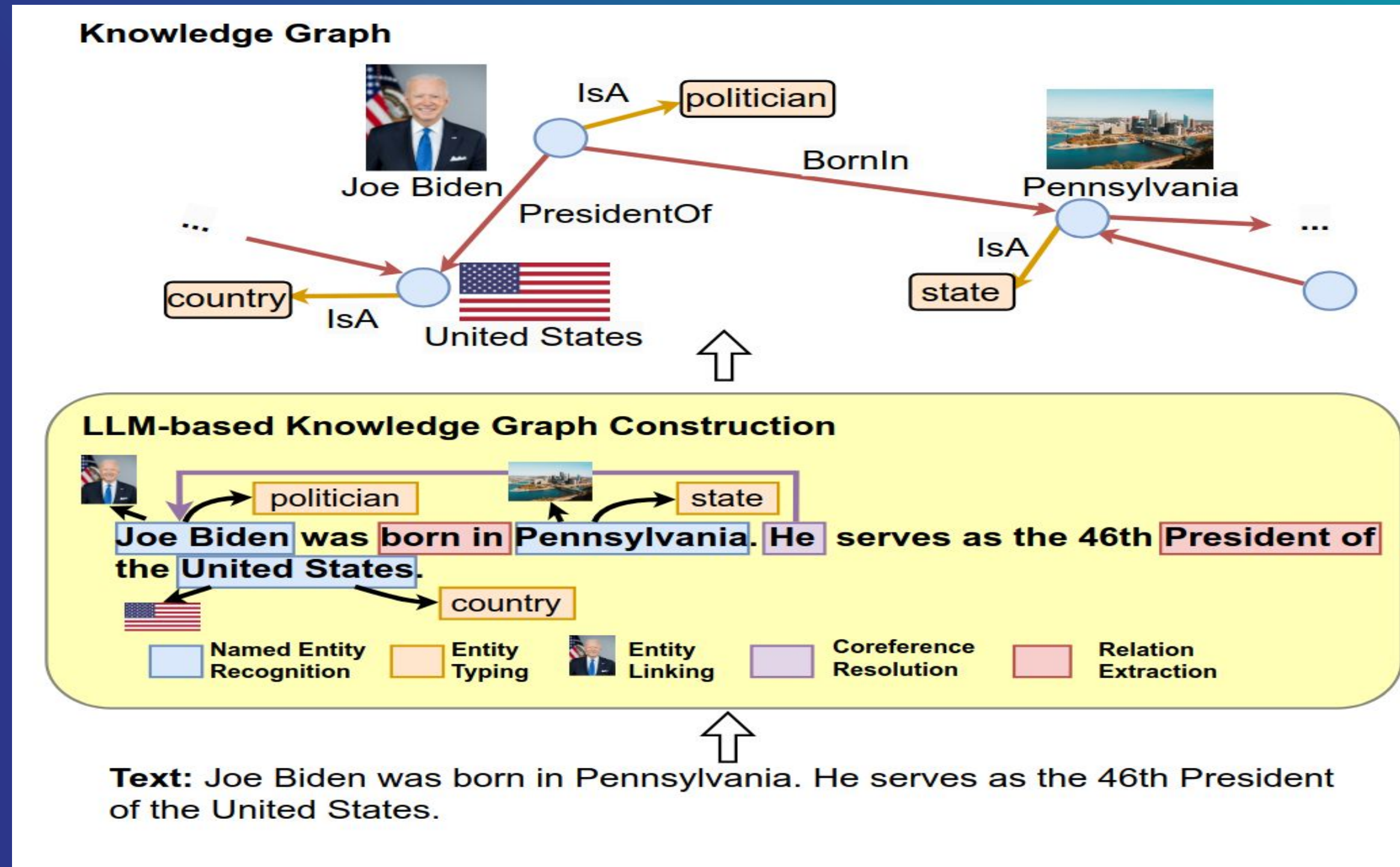
**Retrieval-Augmented Generation (RAG)** advanced retrieval technique, more accurate, contextually aware, and nuanced responses to user queries



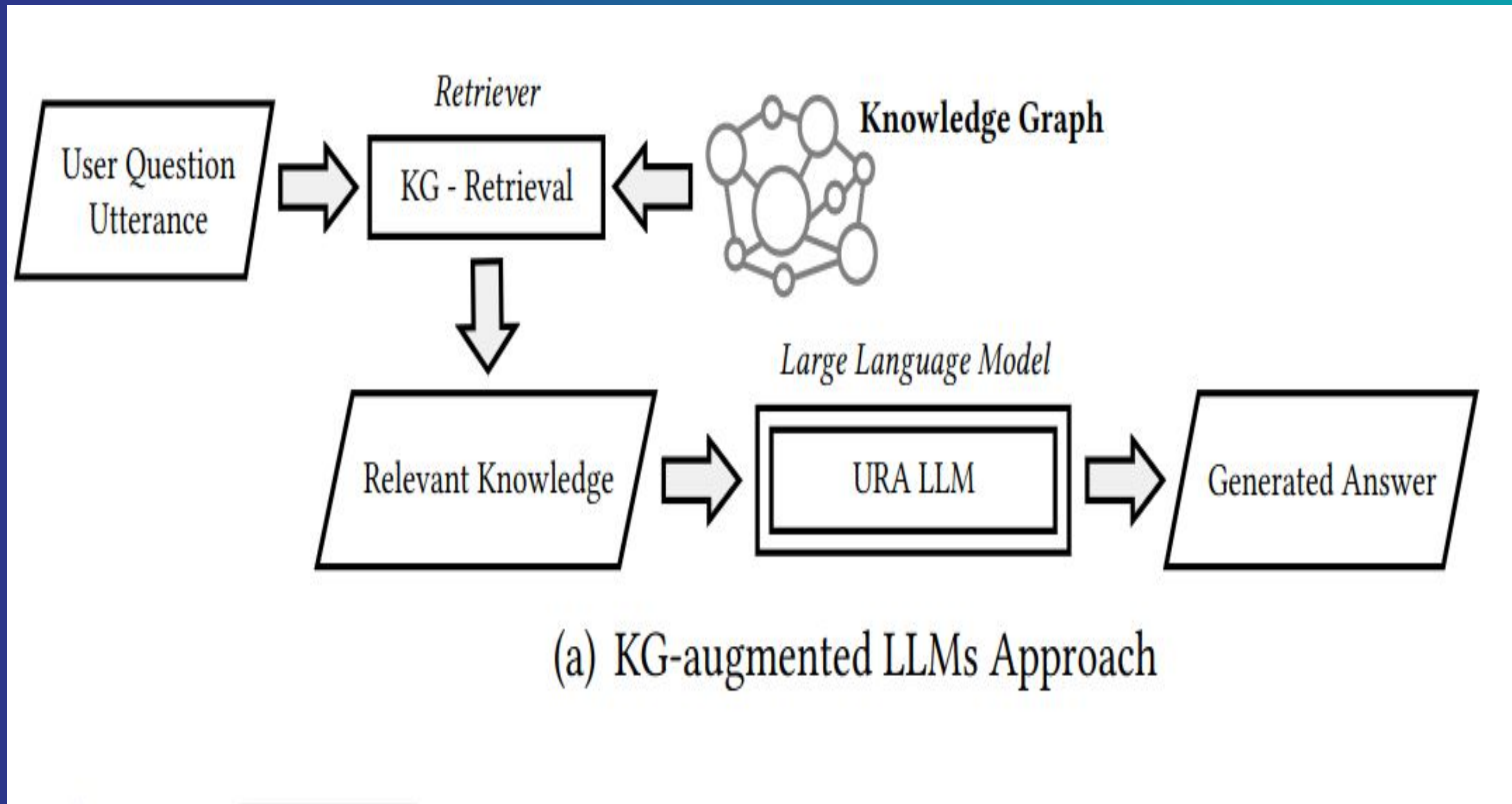


# LLM-augmented Knowledge Graphs

## General framework of LLM-based KG construction



# Knowledge Graphs enhanced LLM generation





# Knowledge Graphs Completion

## Knowledge In Context with GPT (KICGPT) framework

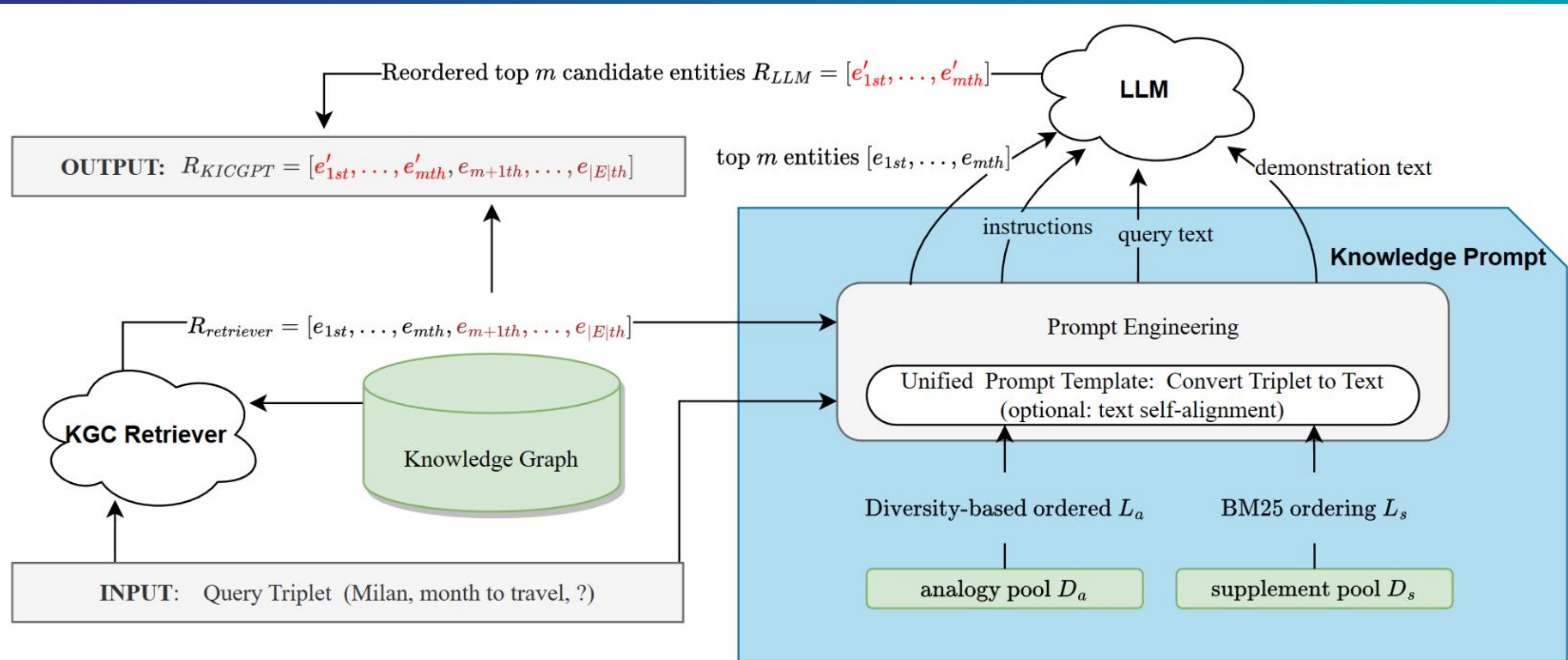
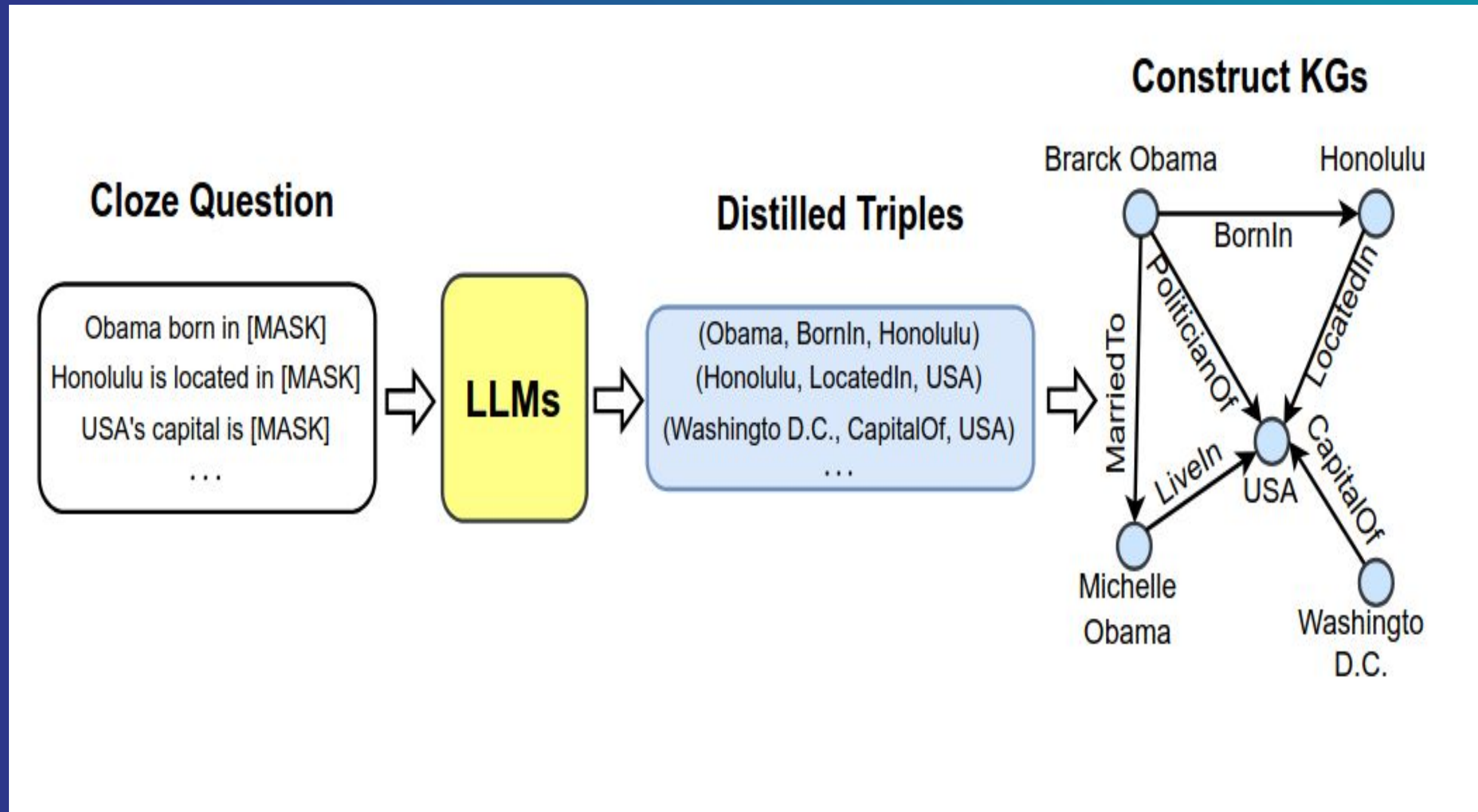


Figure 1: An illustration of the KICGPT framework.

# LLM-augmented Knowledge Graphs

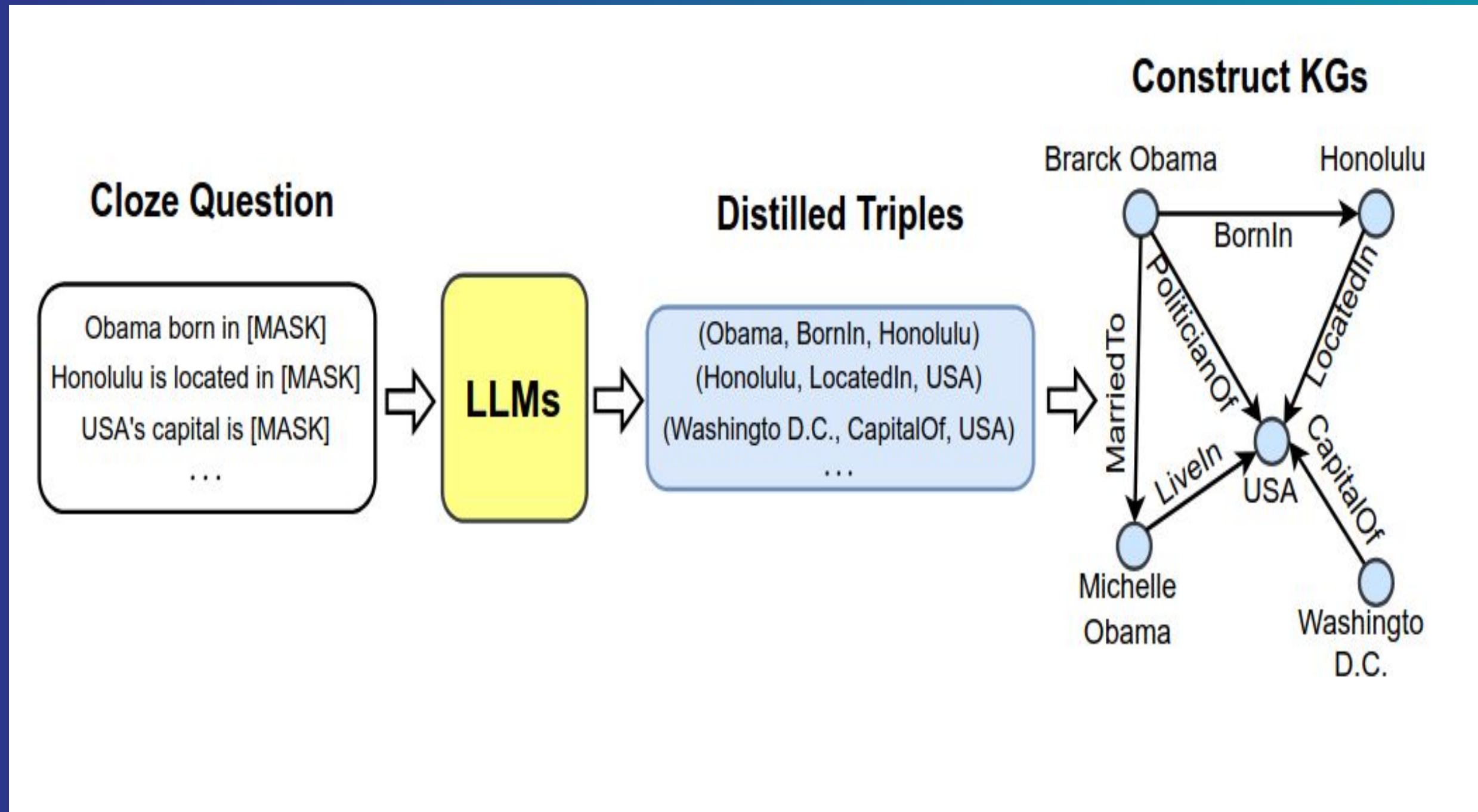
## Distilling KGs from LLMs





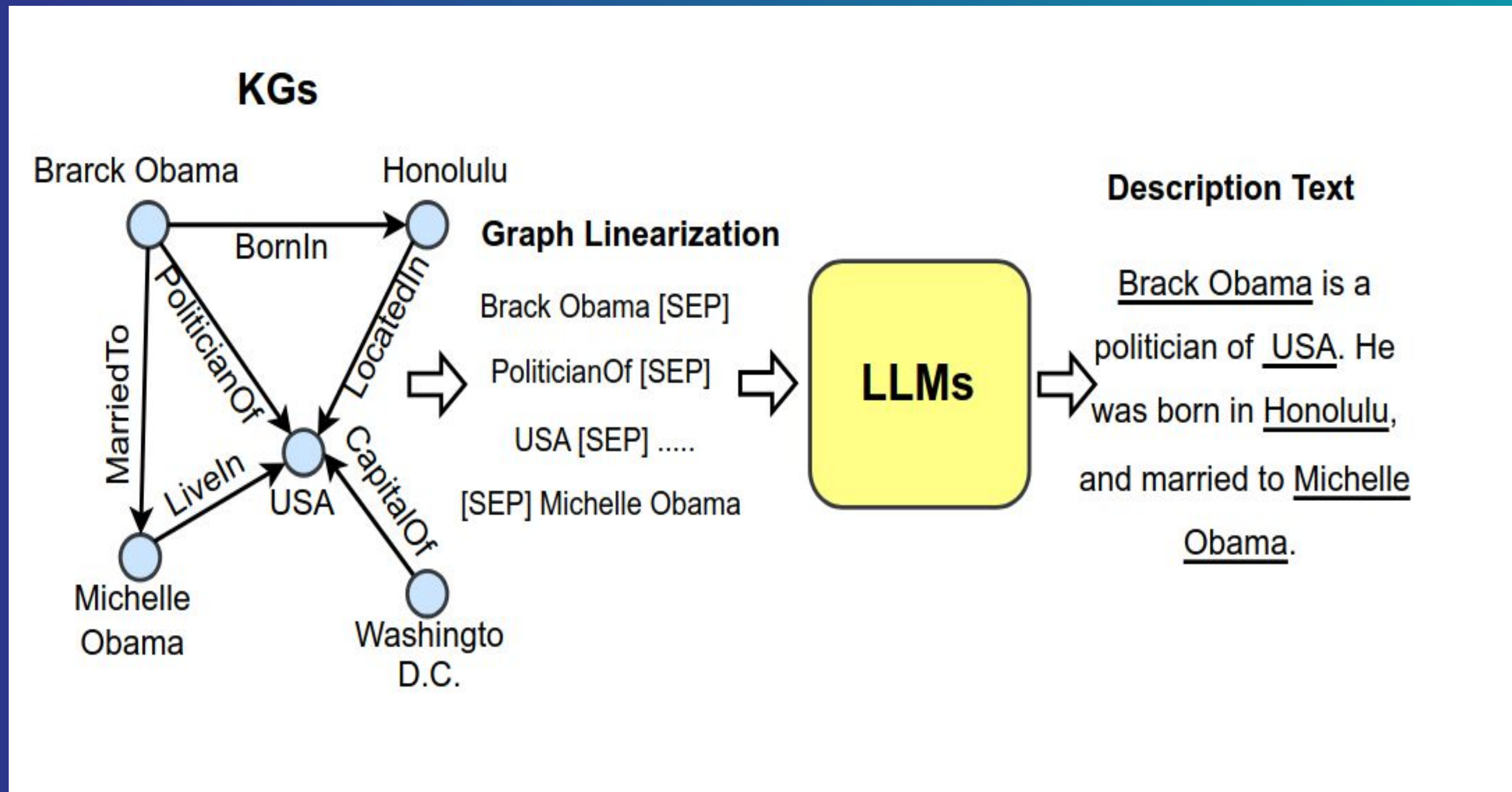
# LLM-augmented Knowledge Graphs

## Distilling KGs from LLMs



# LLM-augmented KG-to-text Generation

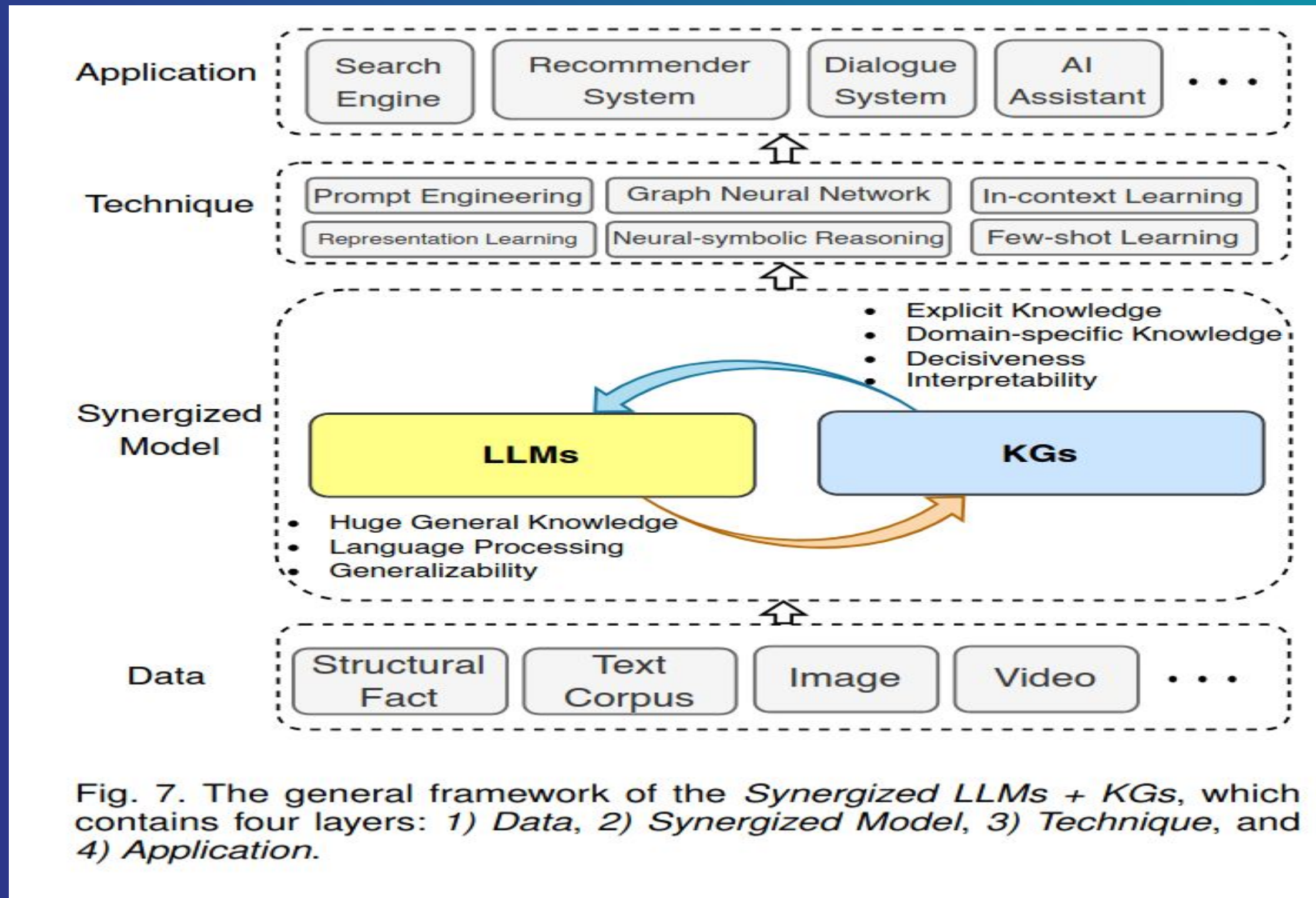
Generate high-quality texts and leverage knowledge from LLMs





# LLMs Synergised with Knowledge Graphs

From perspective of knowledge representation and reasoning



# LLMs Synergised with Knowledge Graphs

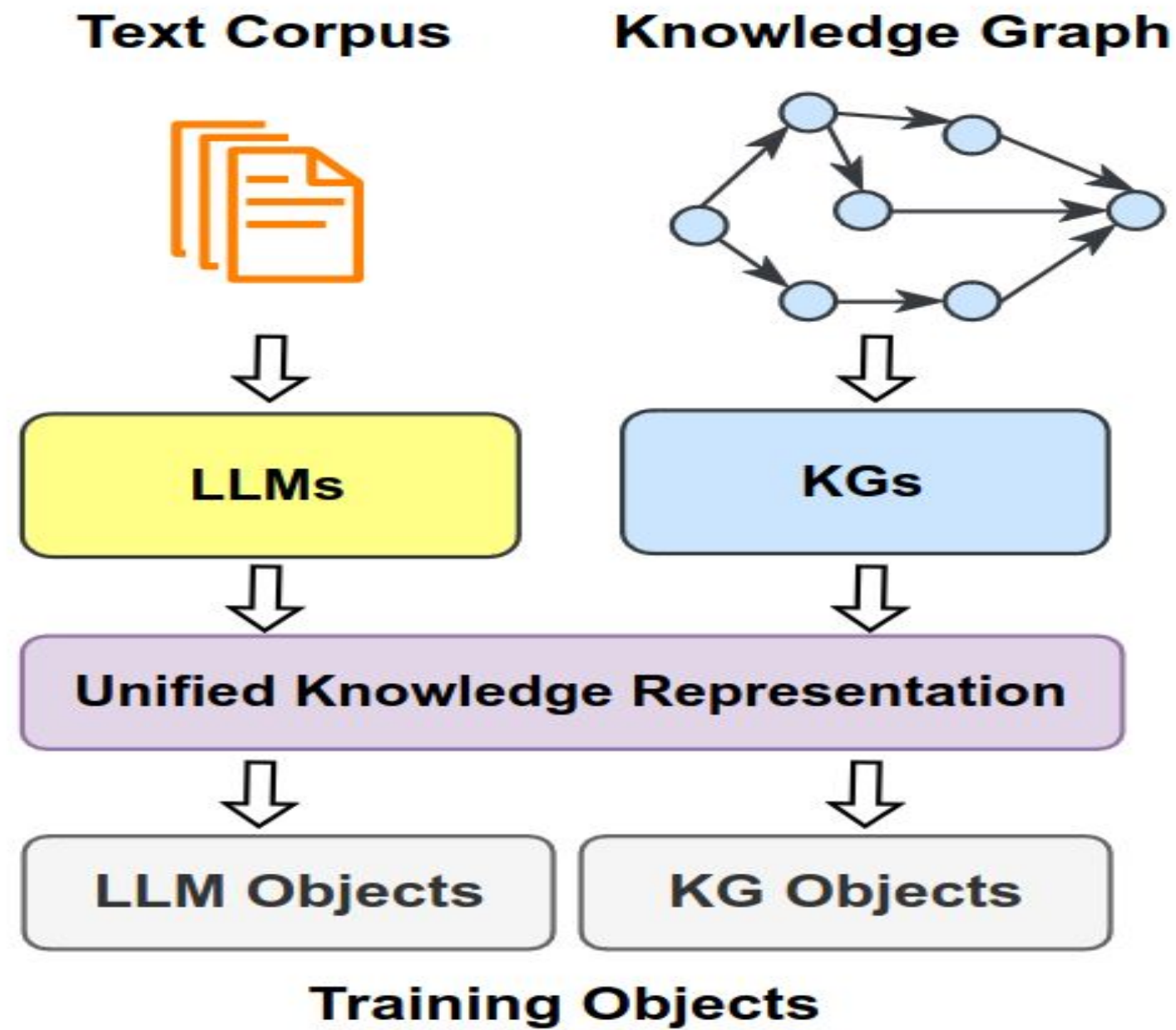
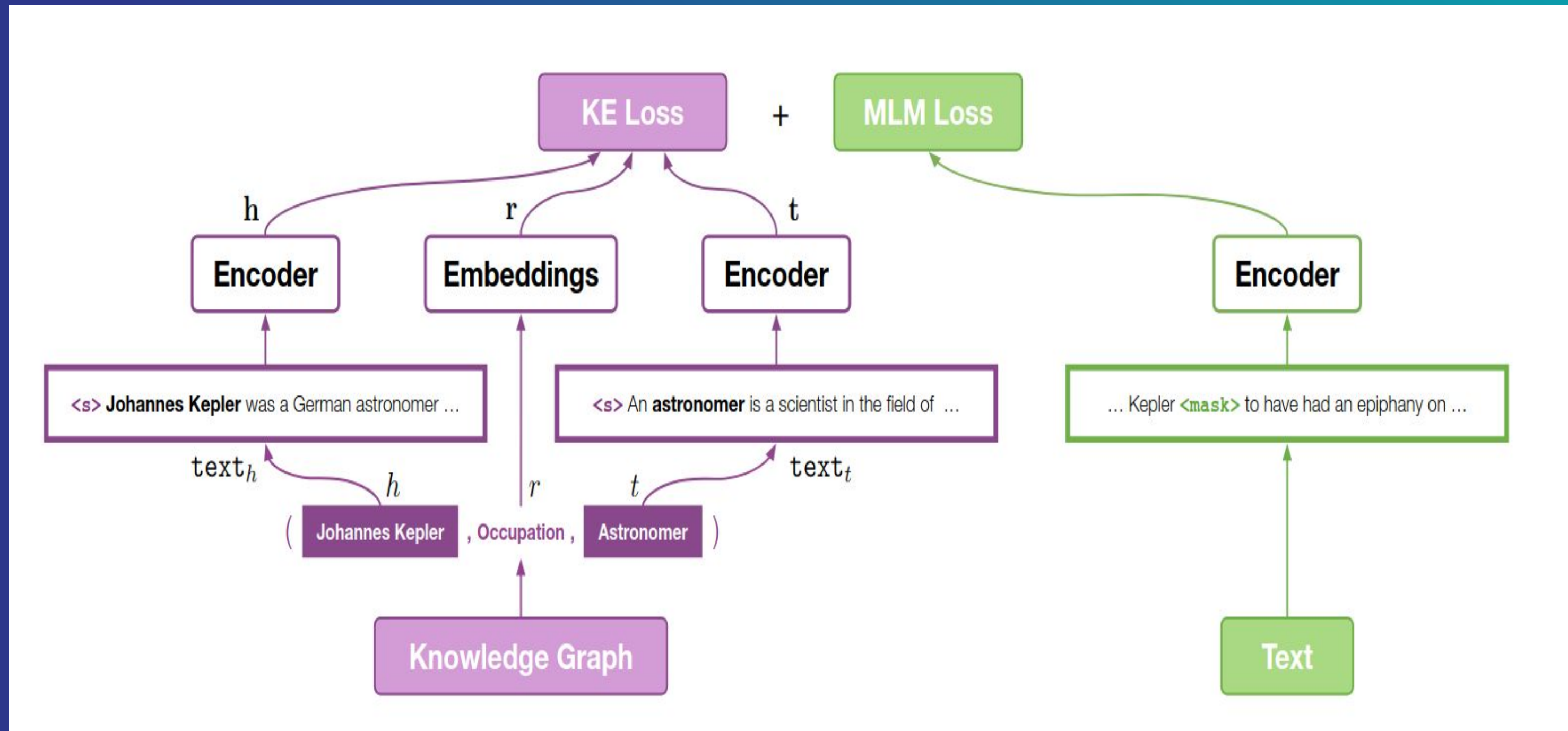


Fig. 25. The general framework of unifying LLMs and KGs for knowledge representation.



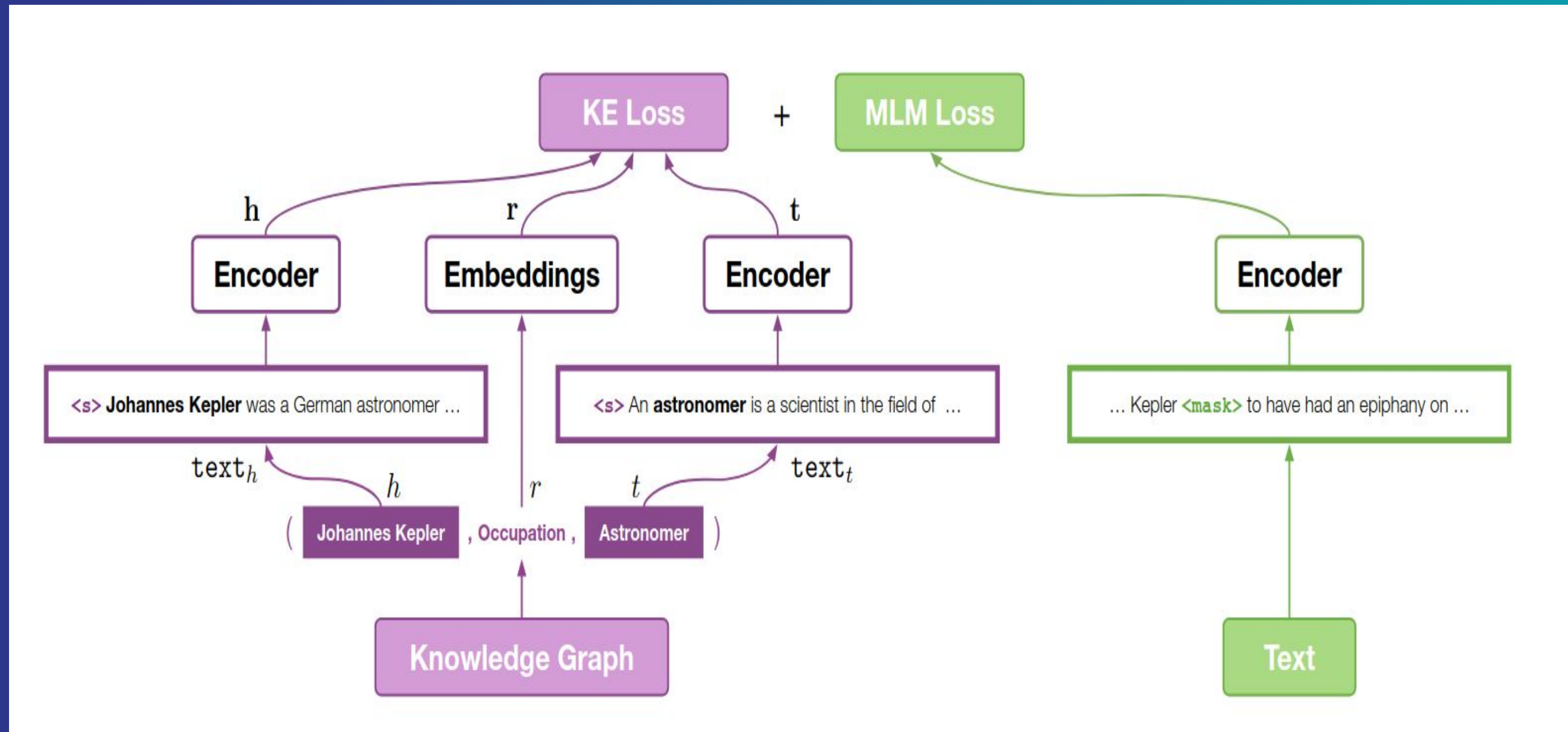
# Unified model for Embedding

## KEPLER: Knowledge Embedding and Pre-trained Language Representation



# Knowledge Graph Compressions

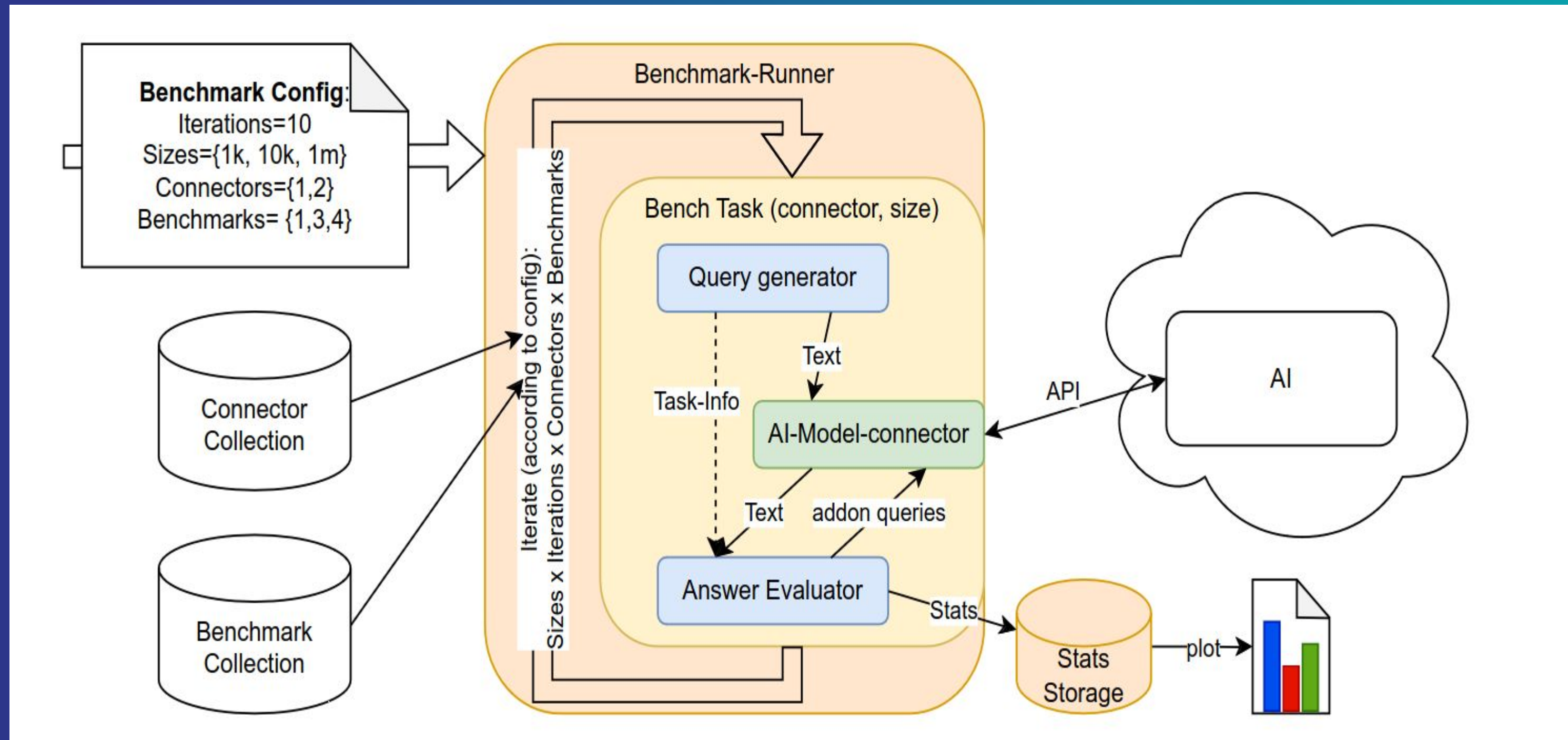
## KEPLER: Knowledge Embedding and Pre-trained Language Representation





# Evaluation and Benchmarking

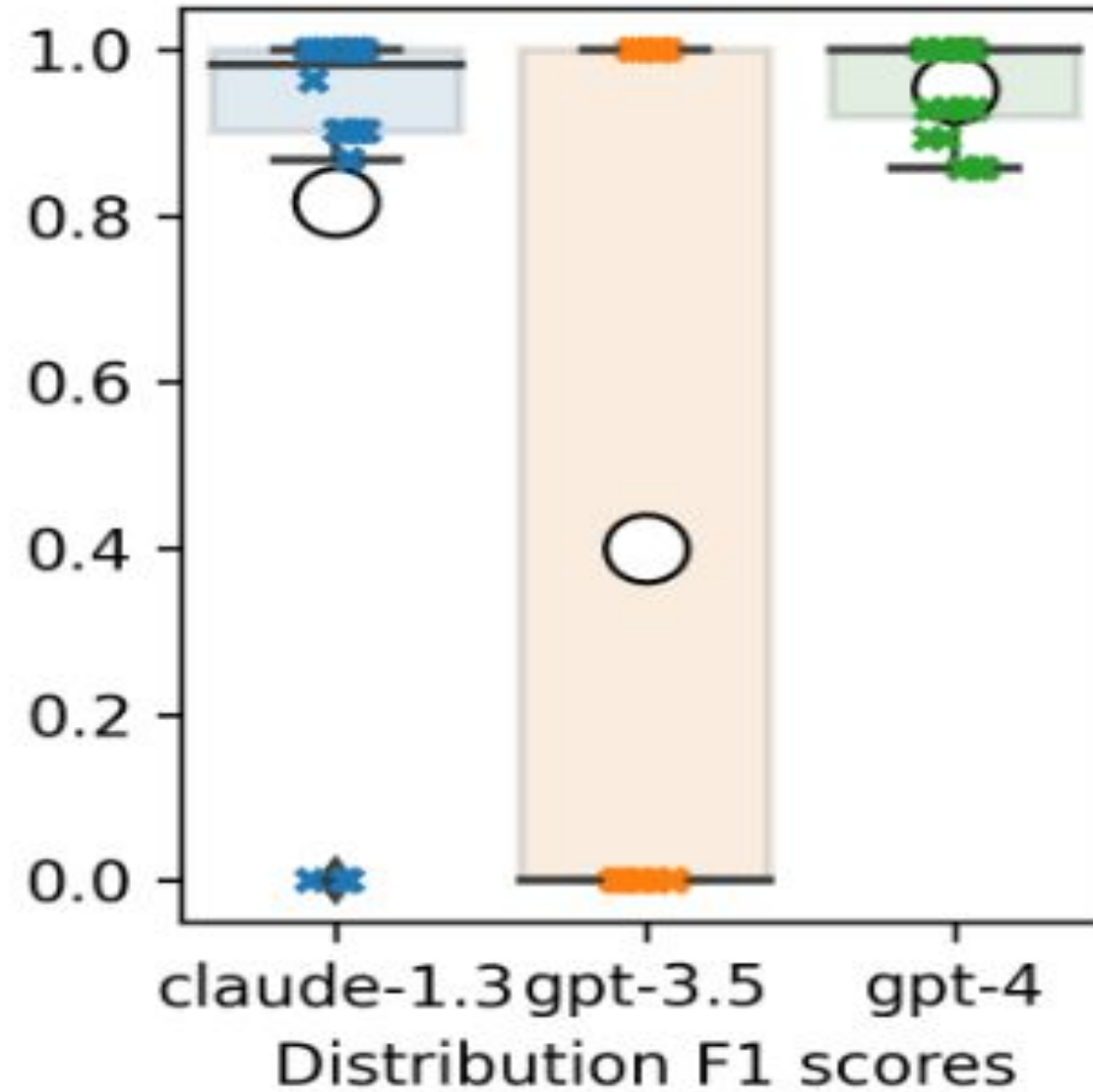
## LLM-KG-Bench: an automated and continuous evaluation platform



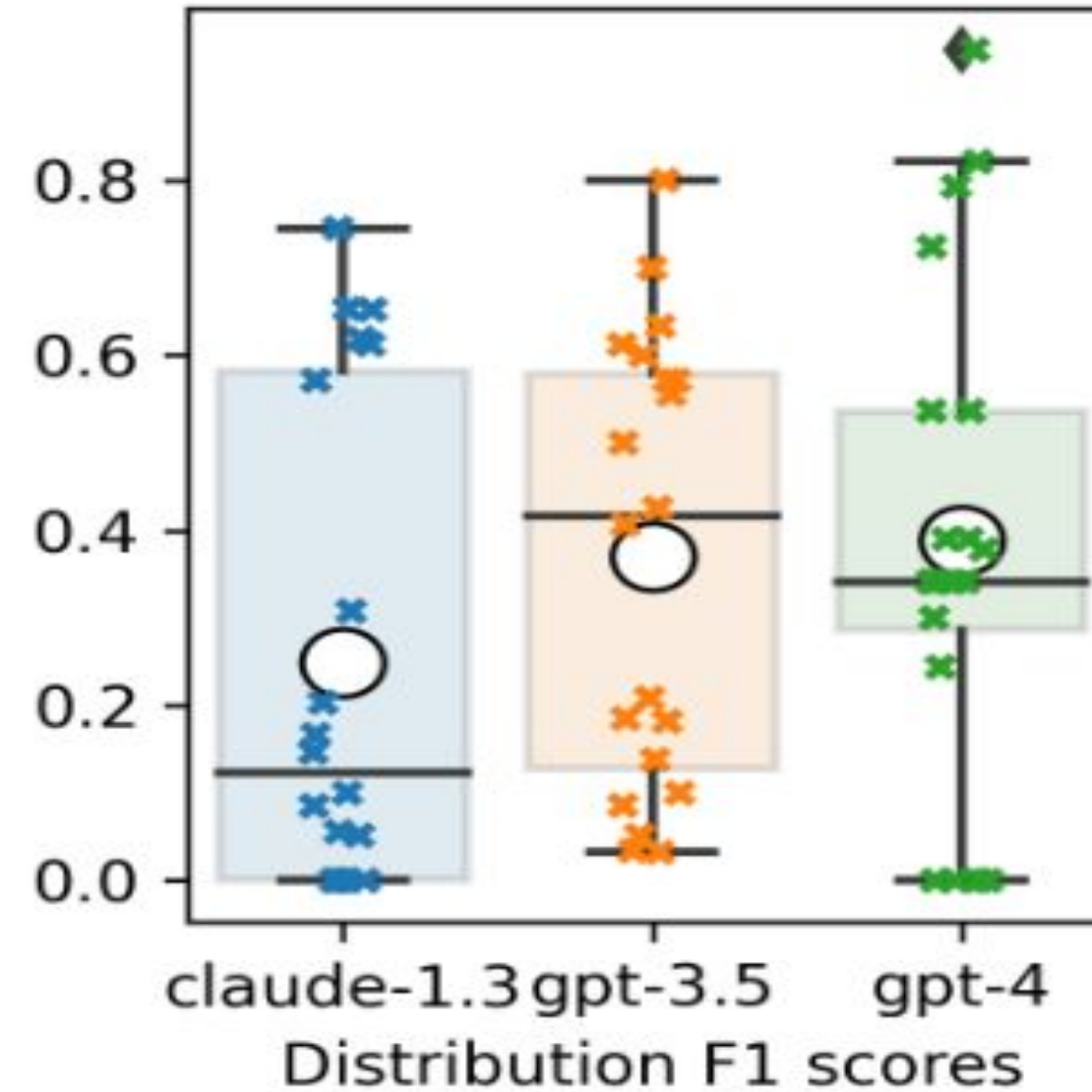
# Evaluation and Benchmarking

Fixing of Errors in Turtle Files

KG Creation from Factsheet Plaintext



(a) Turtle Fixing



(b) Fact Extraction



# Evaluation and Benchmarking

Direct (synthetic data) and indirect (model)

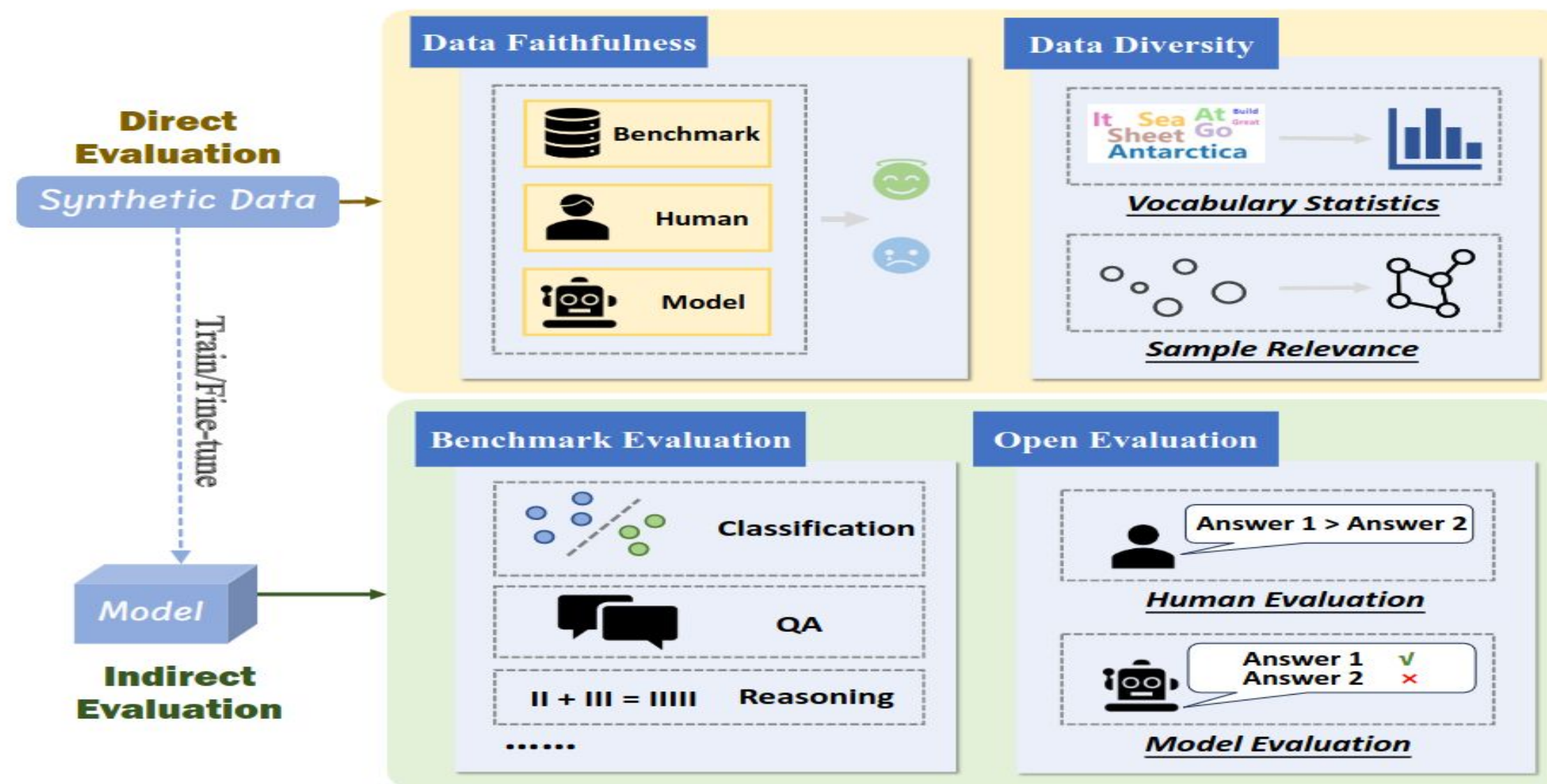
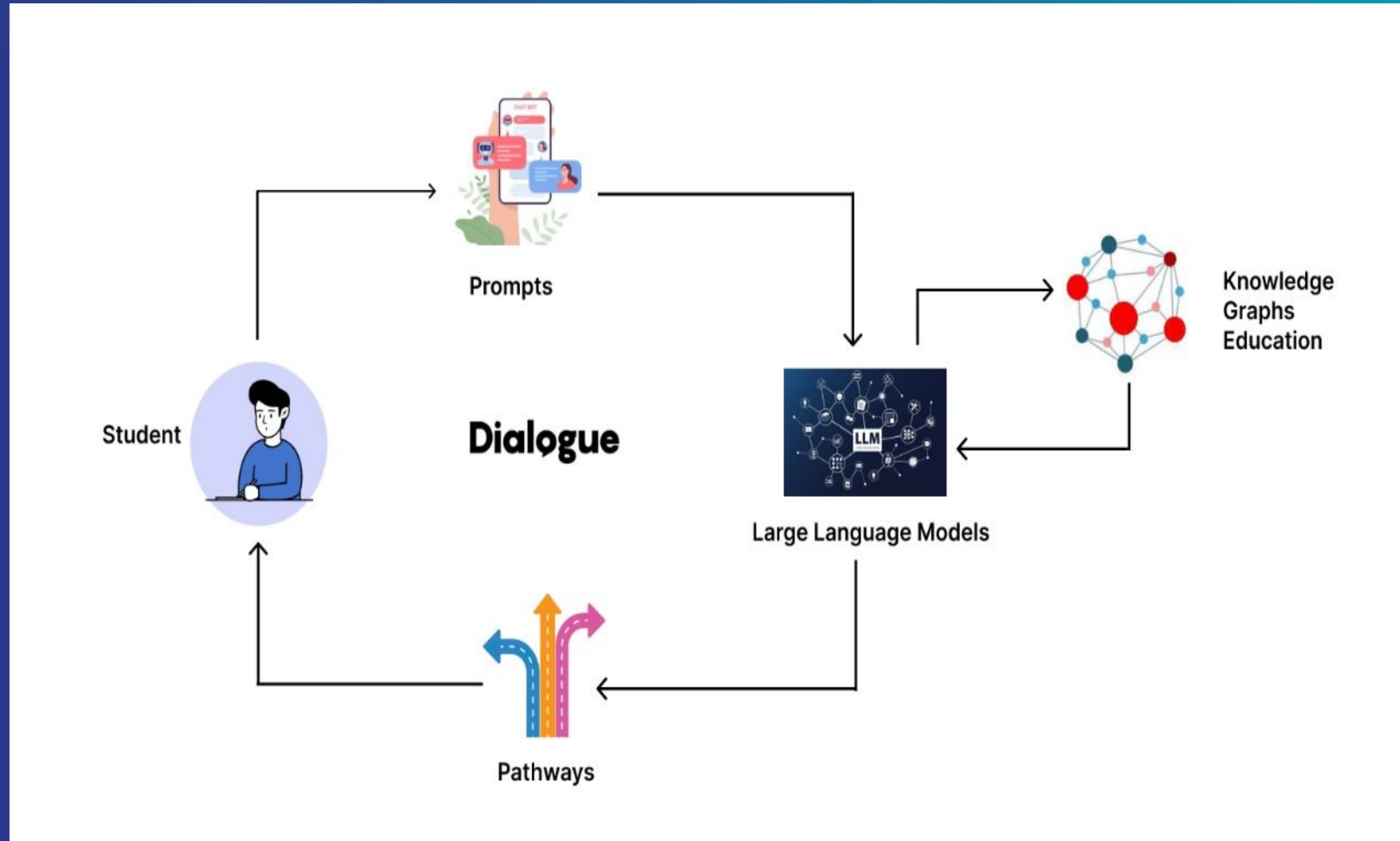


Figure 5: Direct and indirect methods of data evaluation.

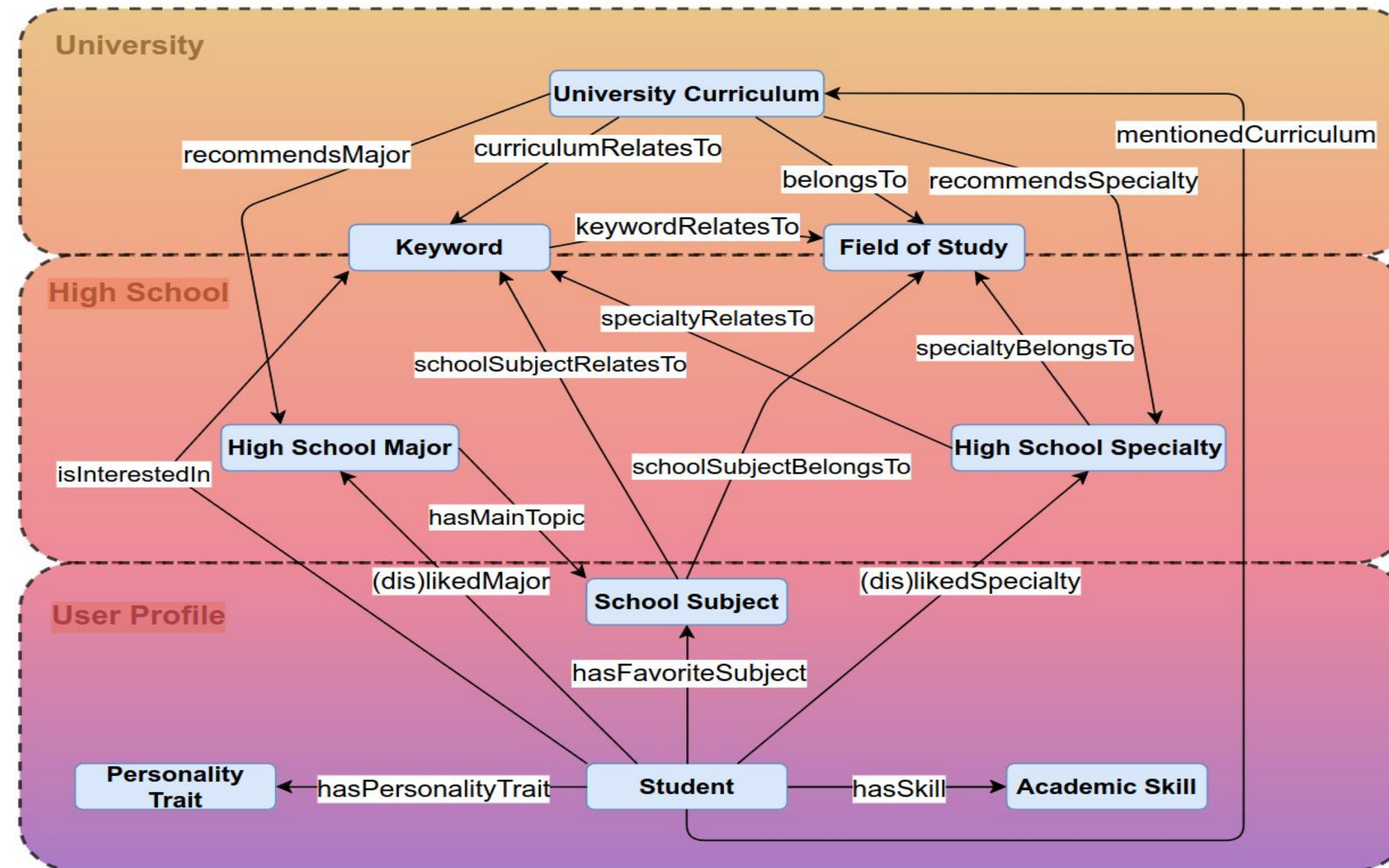
# Generating Personalised Educational Pathways



Using Open-source Language models and Knowledge graphs (Graph RAGs)

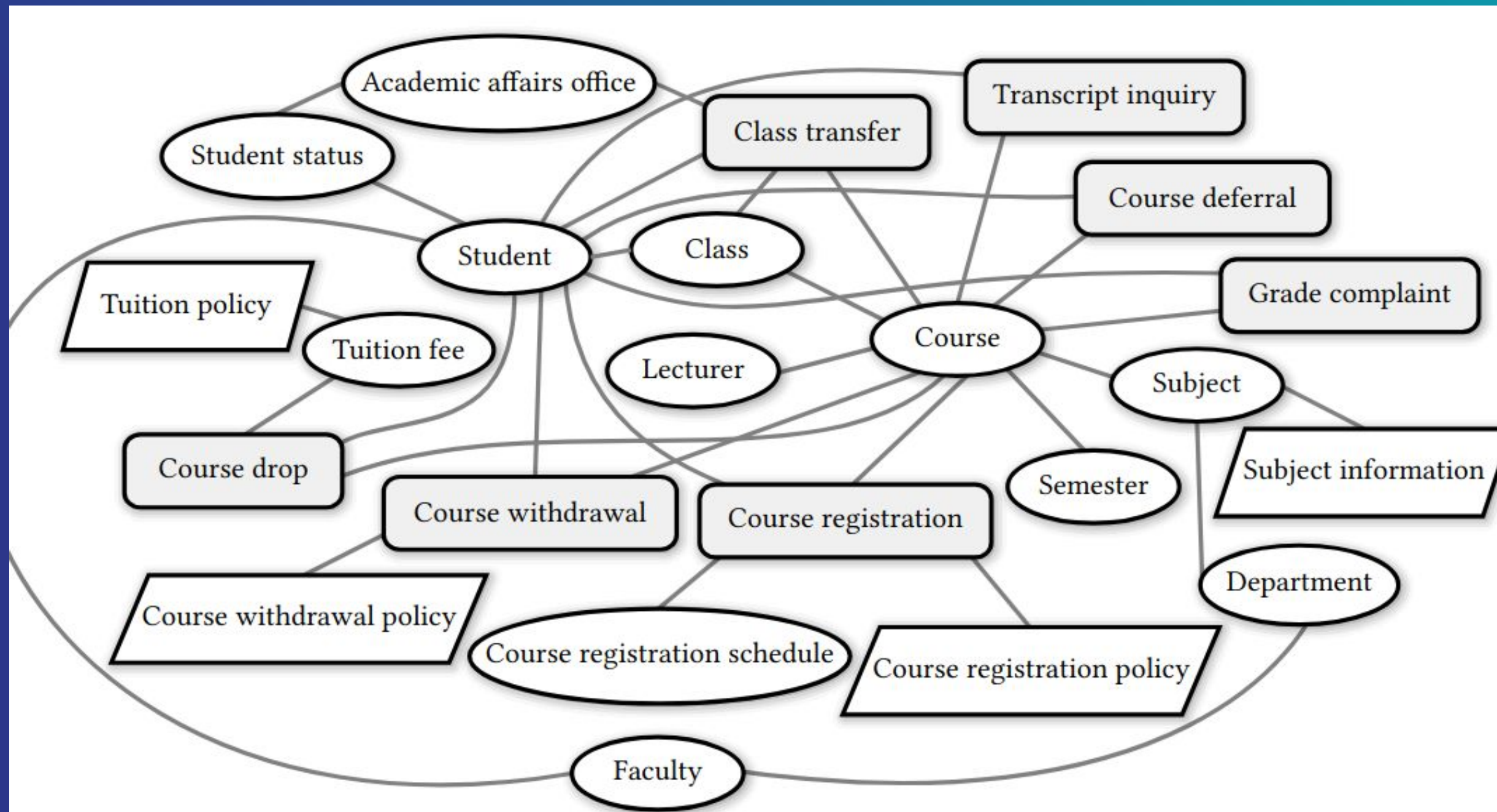


# Education Knowledge Graphs



**Figure 1:** A conceptual overview of EducOnto. The lower, middle and upper parts of the diagram depict the classes and properties related to the User Profile, High School and University, respectively.

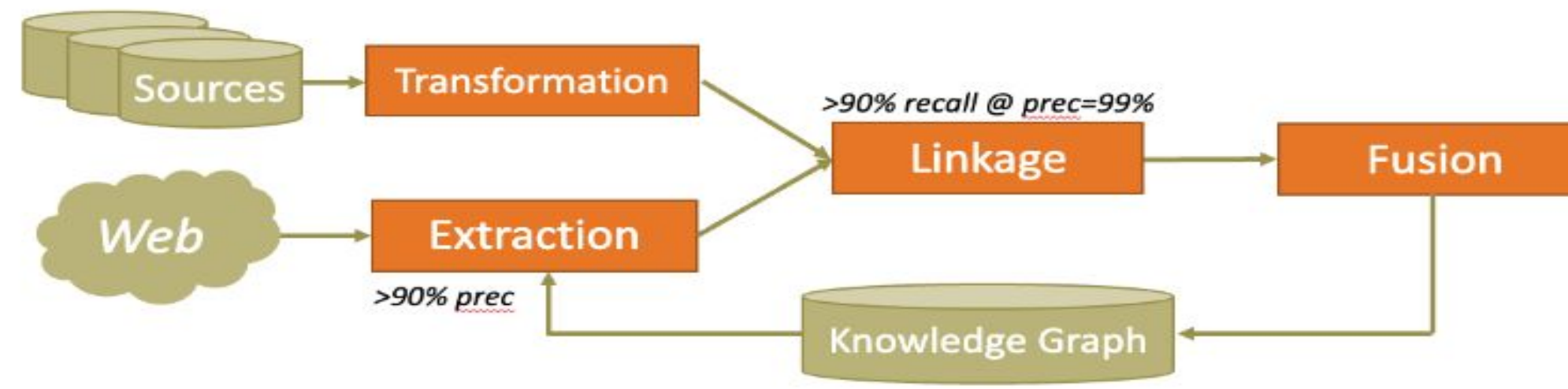
# The E-OED Framework



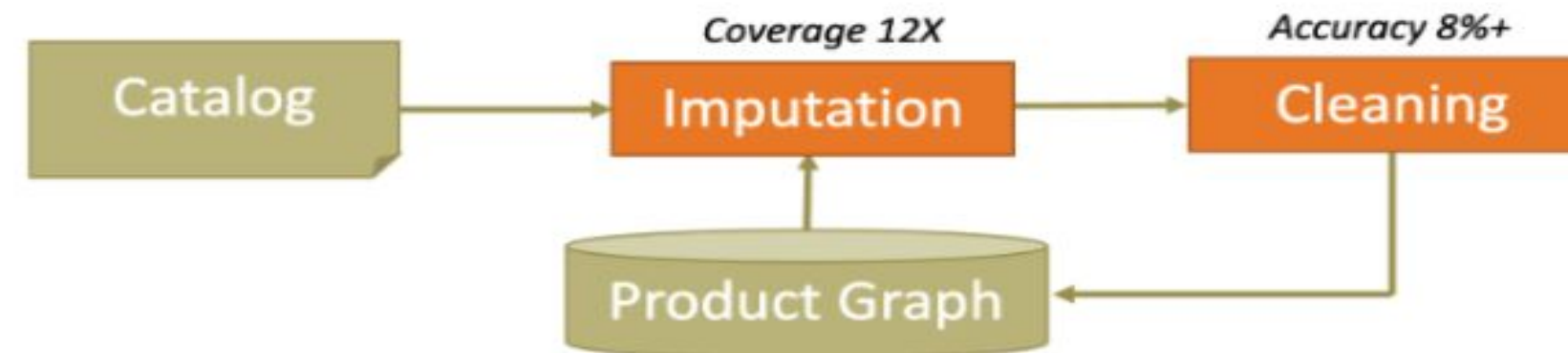
**Fig. 2: Entity type in education domain**



# Generating Knowledge Graphs



(a)



(b)

**Figure 4: (a) Architecture for constructing an entity-based KG. (b) Architecture for constructing a text-rich KG.**

# Generating Knowledge Graphs

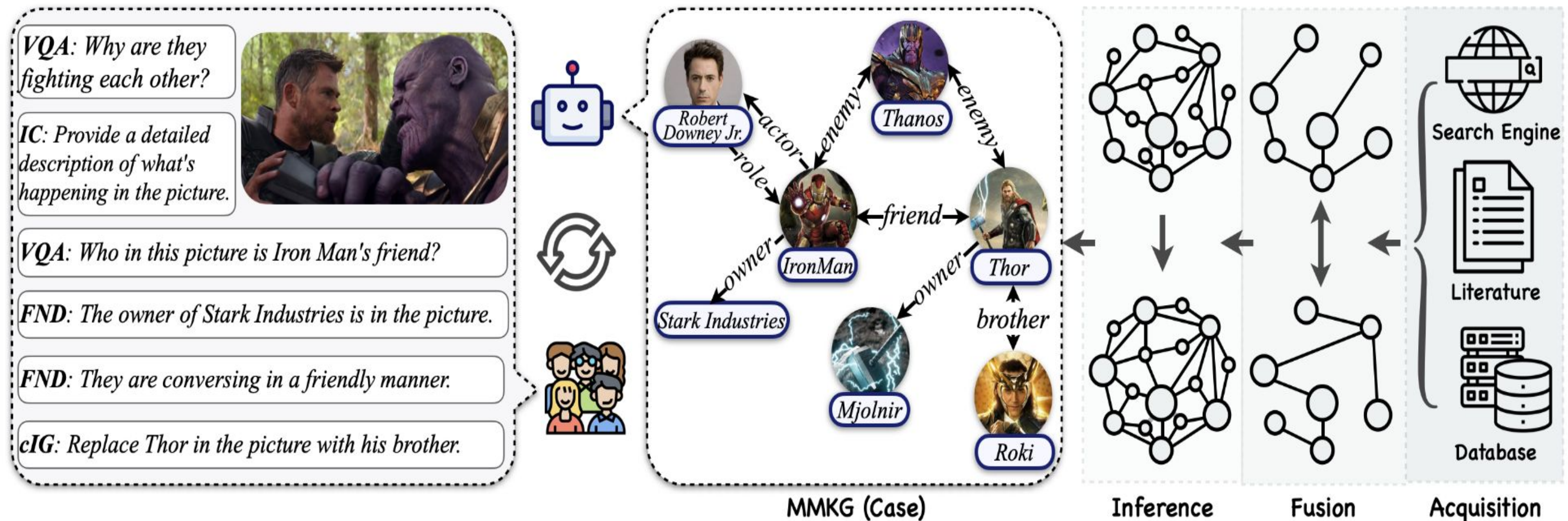
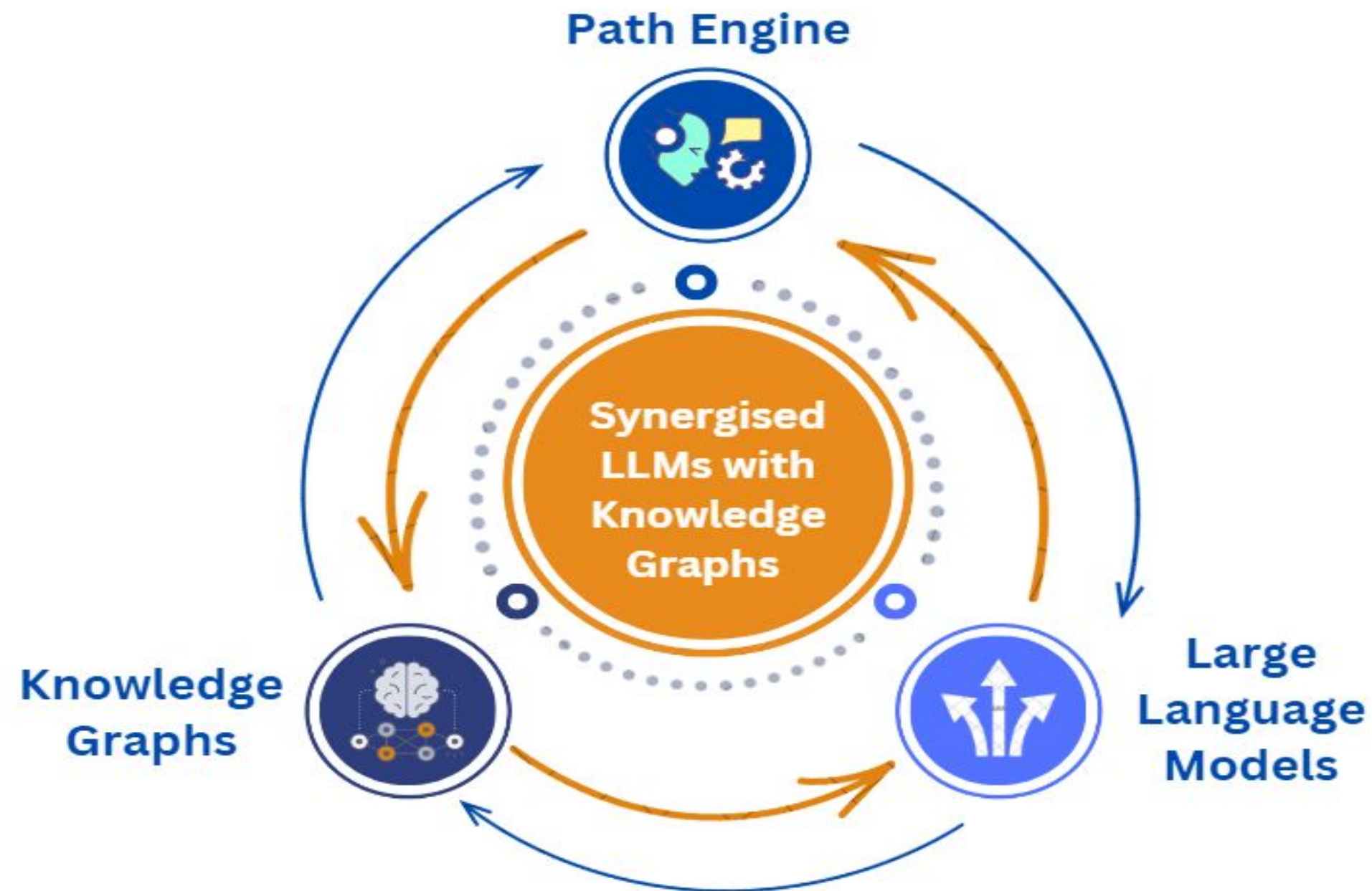


Fig. 3: Roadmap for Multi-Modal Knowledge Graph (MMKG) construction and application in downstream multi-modal tasks.



# Synergised Architecture



perspectives of knowledge representation and reasoning



# LLM HyperGraphs Architecture

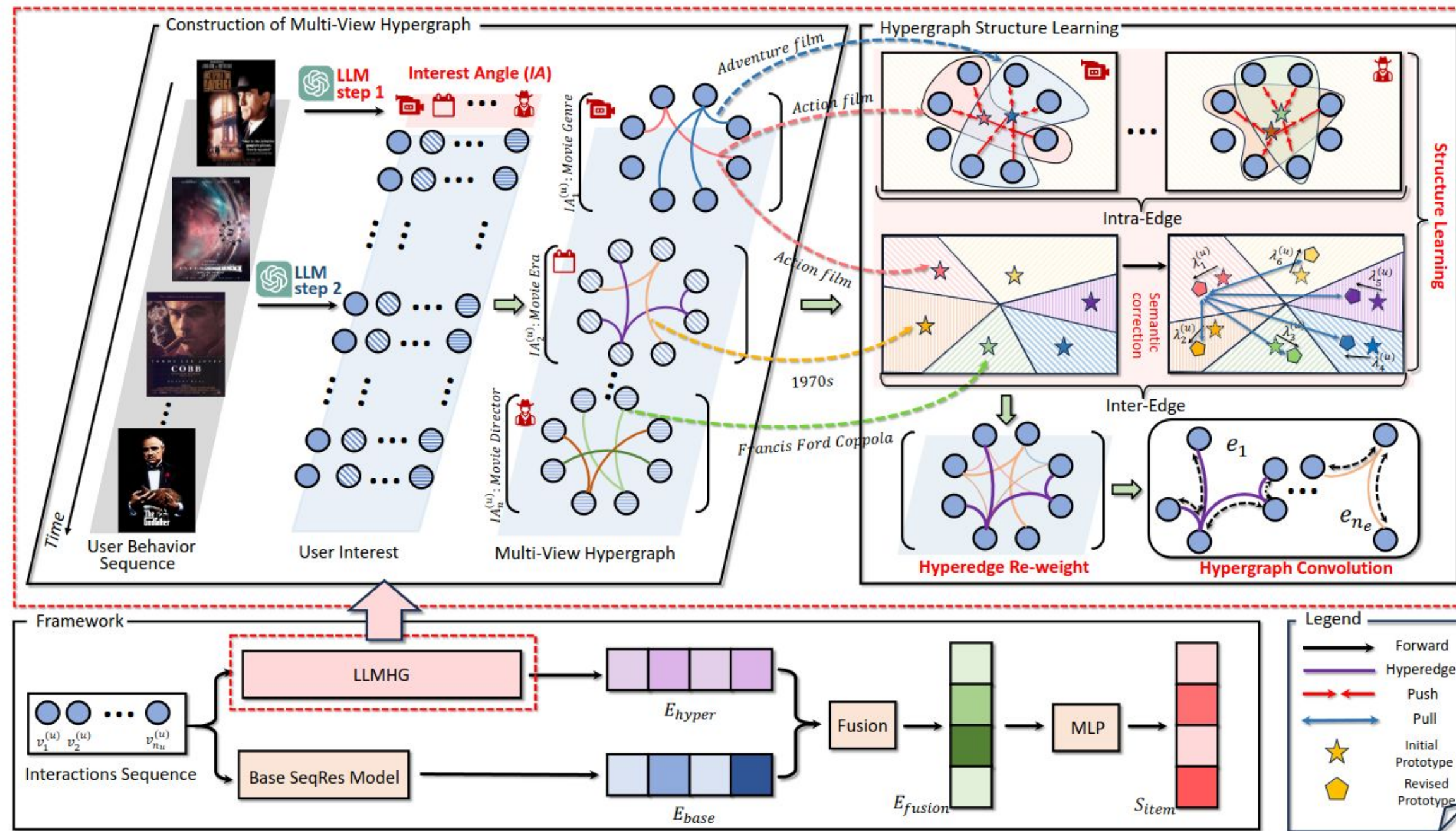


Figure 1: LLMHG includes four major steps: interest angle extraction, construction of a multi-view hypergraph centered on interest angles, hypergraph structure learning for LLM content refinement, and representation fusion for recommendation prediction.



# LLM HyperGraphs Architecture

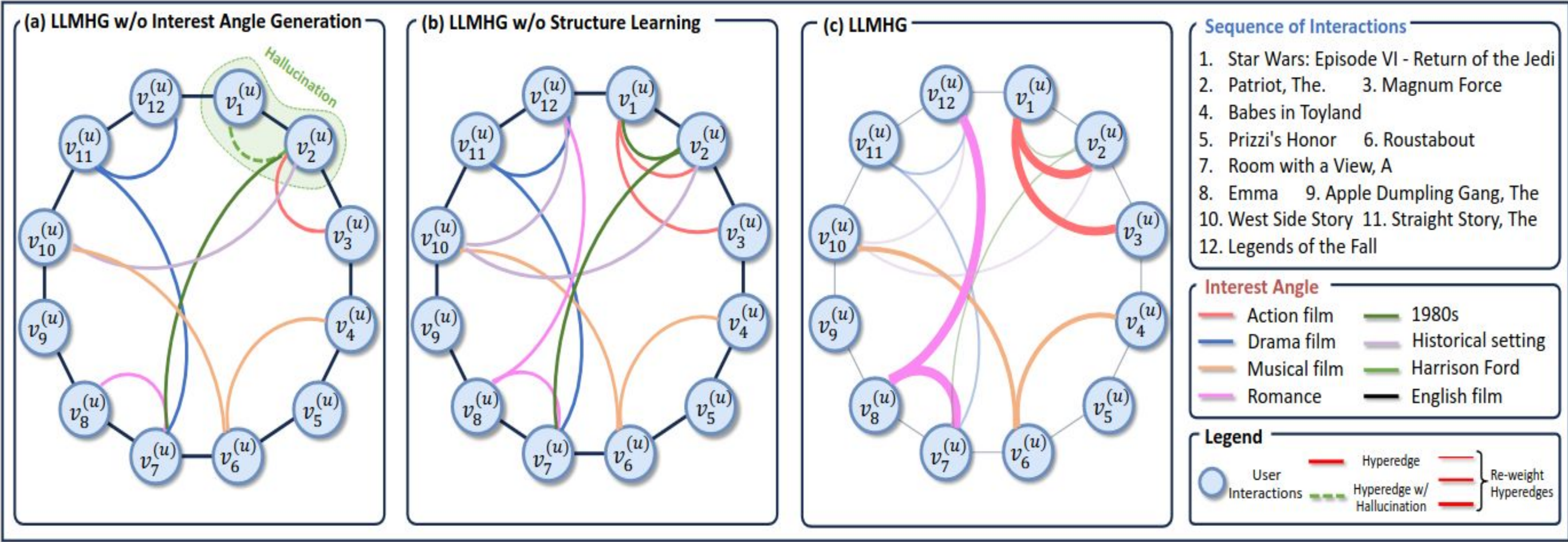


Figure 2: The real case studies (ML-1M) on our (c) LLMHG and ablation models, i.e., (a) LLMHG w/o interest angle generation and (b) LLMHG w/o hypergraph structure learning .

from perspectives of knowledge representation and reasoning



# Graph Compressions

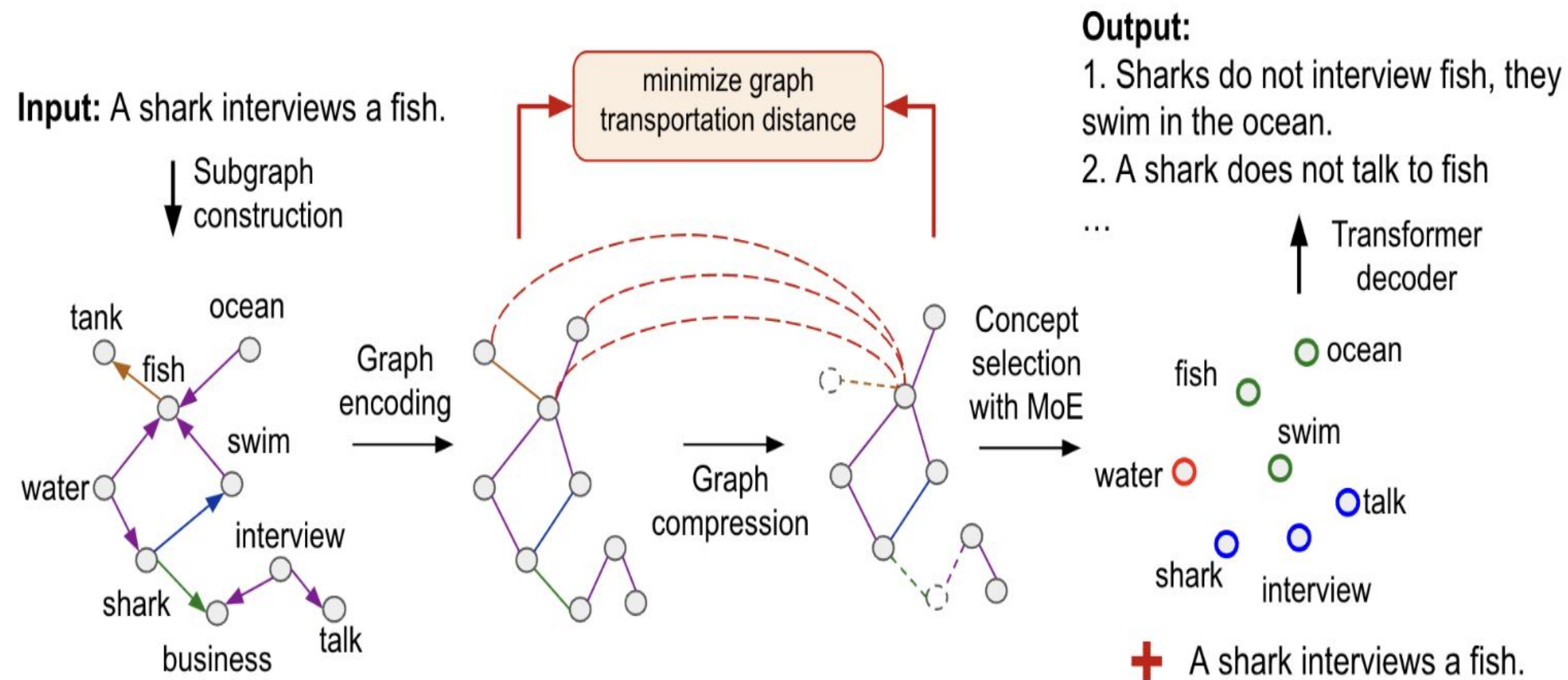
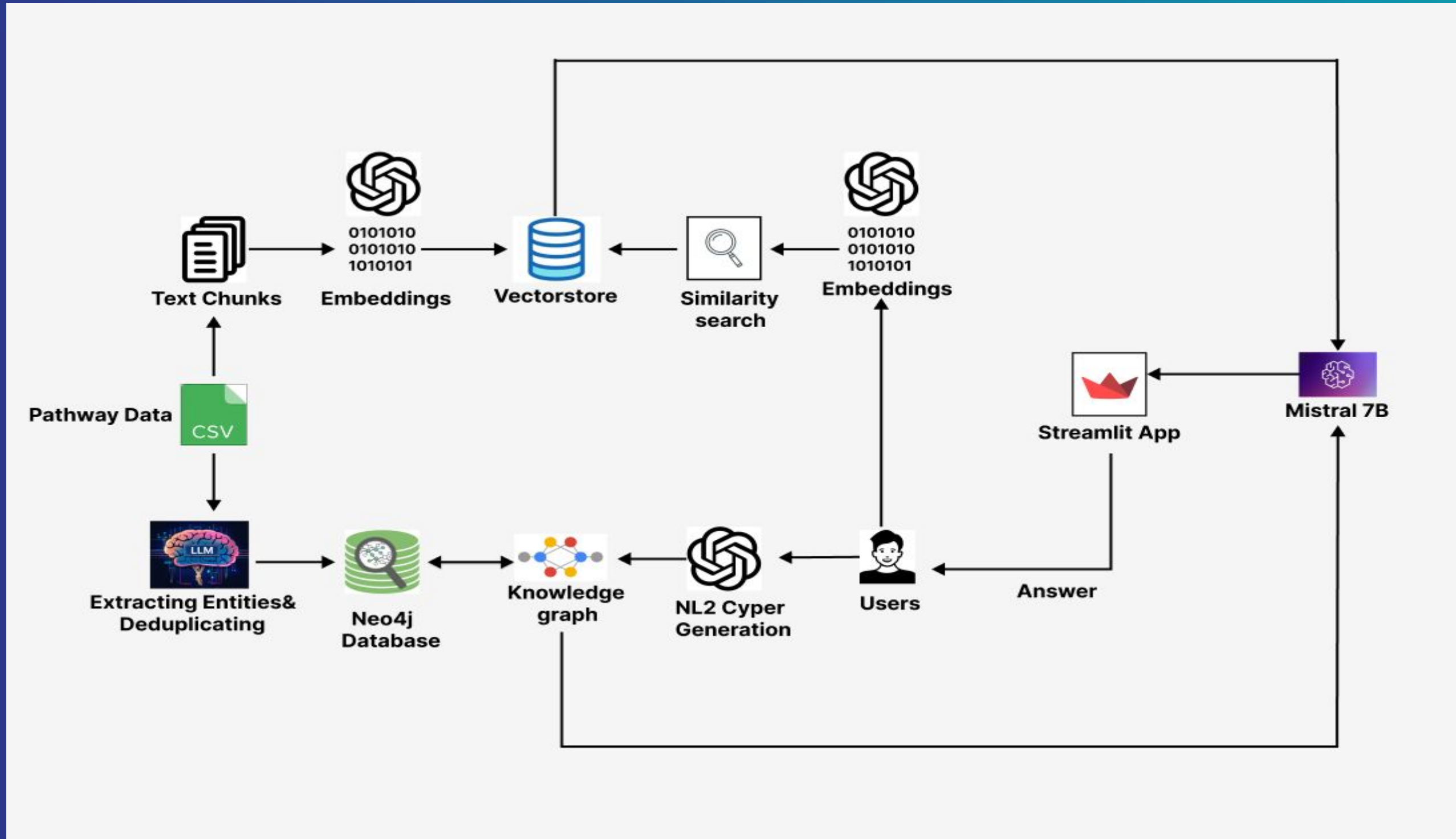


Figure 2: Overview of our approach. We retrieve a subgraph from ConceptNet for the given input sentence, compress it, and use MoE to generate diverse sentences for containing concepts from the compressed graph.

**Node merging (like the Shrink algorithm), Directed Acyclic Graph (DAG) compression, differential encoding to reduce graph size**



# Hybrid Architecture



Combining graph and vector search

# Summary and Outlook

- KGs to ground LLMs and thereby providing contextual responses, limiting hallucinations, and lack of interpretability
- Multi-Modal LLMs for KGs
- Development of sophisticated algorithms to ensure KGs have low-latency to enhance the effectiveness of integrations.
- Synergising LMs and KGs for Birectional Reasoning in context of educational pathways



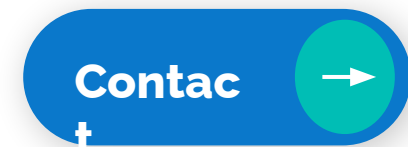
# References

- Special thanks to Jakob Porschmann (Google Berlin)

- Few

References:

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- <https://seucoin.github.io/workshop/llmkg/>
- [https://www.researchgate.net/publication/377318531\\_Unifying\\_Large\\_Language\\_Models\\_and\\_Knowledge\\_Graphs\\_A\\_Roadmap](https://www.researchgate.net/publication/377318531_Unifying_Large_Language_Models_and_Knowledge_Graphs_A_Roadmap)
- <https://github.com/AKSW/LLM-KG-Bench>
- <https://arxiv.org/pdf/1911.06136>
- <https://www.ontotext.com/knowledgehub/fundamentals/what-is-graph-rag/>



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**Githb:**    **[https://github.com/adytiaa/naavi\\_os](https://github.com/adytiaa/naavi_os)**



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